

/// Component, Fault, and Event Data Dictionary

I-ETMS[®] Protect

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/// Revision History

Date	Revision	Description	Author
	0.1	This data dictionary is based off revision 3.1 of WCR-SYR-1058	
12/1/2022	1.0	DCR 174896 - Initial revision for I-ETMS Protect DCR 172626 (154300, 173036, 162381, 173783) ports from 6.3.24.0 DCR 169095 - Added 0x067A DCR 157699 - added configuration codes 90-93 DCR 177662 - Added 0x0619	Wabtec
12/7/2022	1.1	DCR 179791 (171260) - modified disengagement scenarios	Wabtec
4/23/2024	1.2	DCR 178626 - HOTCC Interface DCR 175499 - Rear End Position Adjustment from PTC-EOT impacts on Automatic Rollup of Authorities and Train Integrity DCR 221221 - Clarify Front BPP comparison logic in Sensor Mgmt White Paper DCR 223954 - EOT Fault Modifications DCR 234614 (231660/230698) - Added Event Codes and Disengagement Scenarios	Wabtec
2/11/2025	1.3	DCR 251927 - Added GPS faults 0x05E0 and 0x05E1 DCR 259455 - Added Enhanced Fault Diagnostic Code D32 and Fault 0x0808 DCR 266969 (265440) - Added Fault Codes	Wabtec
8/20/2026	1.4	DCR 293796 (216235) - Added Fault Code 0x070A DCR 292755 (177310) - Modified Diagnostic Code D10 DCR 293561 (129052) - Added Fault codes 0x0819 and 0x081A. Modified Faults 0x0702 and 0x0800 DCR 292758 (175475) - Changed control type for D07 from TMC DIO to LIG DCR 293317 (167548) - Update system response for faults 0x0741 and 0x0743 DCR 261485 (167739) - removed faults 0x0744 and 0x0745 and updated fault 0x0620 DCR 294305 (208184) - Adding new component for EBI2 DCR 294215 (176582) - Adding Event Code for power cycle DCR 294027 (159860) - Adding support for the SR KSS EMU DCR 258516 - Added Fault 0x085D DCR 324803 (234555) - Added Fault 0x0518 DCR 324222 (219729) - Removed CRM DMU from Faults 0x050F and 0x0510 DCR 294891 (170115) - Updated System Response for Fault 0x050F DCR 247470 (163637) - Updated Fault 0x0790 DCR 294541 (143852) - Updated System Response for Fault 0x0674 DCR 294881 (165452) - Updated Condition to Clear for Fault 0x0711 DCR 294887 (165868) - Updated (02088) Message Definition DCR 294890 (168521) - Updated System Response for Faults 0x0703, 0x0704, and 0x0705 DCR 294906 (174637) - Modified 0x0707 Fault so	Wabtec

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Date	Revision	Description	Author
		that it does not prevent initialization DCR 305112 (177311) - Remove D23 DCR 305963 (184783) - Remove D14 and D15 DCR 321769 (196133) - Update 0x0800 and 0x0702 Conditions to Trigger DCR 324778 (228481) - Update Configuration Codes 83-89 DCR 324840 (240763) - Update 0x0708 fault to ensure Init key is not displayed DCR 264137 - Add new faults to support Canada DCR 385232 - Add Fault Code D34 DCR 350091 (340098) - Updated Faults 0x0115, 0x0116 and 0x0117 DCR 350090 (340216) - Added Diagnostic Code D33 DCR 350089 (340244) - Updates to Configuration Code to Component Code Mapping Table DCR 391059 - Update to Fault 0x0518 DCR 294864 (160329) - Add new Event Codes DCR 386286 (157094) - Add D-Code fault and event Codes to 2087 Fault Report Message content DCR 395791 - Format Updates DCR 294380 (281394) - Add BPP2 as a third source for LIG vehicle types DCR 324944 (248583) - Keep faults 0x0101, 0x0104, 0x0107, and 0x010A suppressed for LIG locomotives DCR 294578 (155030) - Fault 0x0692 should not apply to SR FLRT3 DMU DCR 396987 - Fault name correction for 0x053E DCR 404175 - Update Fault 0x074A	

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/// 1 Introduction

1.1 Purpose

Support of the messages identified in the 'Positive Train Control Office – Locomotive Segment Interface Control Document (ICD)' document.

Definition of enhanced diagnostic codes displayed on the on-board segment screen.

Definition of event codes and disengagement scenarios used to analyze the performance of the on-board segment.

1.2 Scope

This document defines the Wabtec vendor code, I-ETMS system code, I-ETMS component codes, I-ETMS configuration codes, I-ETMS fault codes, I-ETMS event codes, and I-ETMS disengagement scenarios.

1.3 Time Stamp

When each fault in this document is logged it is time stamped with time based on Universal Time Coordinated (UTC). The system uses GPS for time correction.

1.4 Reference

The following document is referenced by this document:

- I-ETMS® Protect PTC Office-Locomotive Segment Interface Control Document IV132.0, WEG-ICD-11097. Wabtec.
(See *I-ETMS Protect Compatibility and Compliance Matrix for applicable version*)

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/// 2 Wabtec Vendor Code

L2I380

Vendor Code
WRE

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/// 3 I-ETMS System Code

L2I242277

System Code
PTC

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/// 4 I-ETMS Component Codes

L2I378

Component ID	Component Name	Description
1	CPU1	Train control processor 1 in the TMC
2	CPU2	Train control processor 2 in the TMC
3	CPU3	Train control processor 3 in the TMC
4	CPU4	Energy management processor in the TMC
5	EBI	ETMS Brake Interface card in the TMC
6	RSM	Router/Switch Module in the TMC
7	IOC	Input/Output Concentrator card in the TMC
8	DIO	Discrete Input/Output card in the TMC
9	EBI2	ETMS Brake Interface card in the TMC
10	CPU10	ITCM processor in the TMC
11	CDU1	Primary Cab Display Unit
12	CDU2	Secondary Cab Display Unit
13	LOCO_ID	Locomotive Identification Module
14	Config File	On-board Configuration Files
15	CHM	Crash Hardened Memory
16	CSM	Cab Signal Module
17	Discrete Interface	Locomotive Discrete Interface Cables
18	DP	Distributed Power
19	EAB	Electronic Airbrake
20	EOT	End of Train
21	RESERVED	<i>Not Currently Used</i>
22	ILC	Integrated Locomotive Computer
23	LIG	Locomotive Interface Gateway
24	Tamper Switch	TMC Tamper Switch
25	TMC	Train Management Computer
26	Track Data	Track Database
27	Wheel Tach	Wheel Tachometer
28	AG	Application Gateway
29	EM	Energy Management
30	Locomotive	Locomotive
31	GPS1	Global Positioning Satellite Receiver 1
32	GPS2	Global Positioning Satellite Receiver 2
33	HOTCC	Head of Train Communication Controller

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/// 5 I-ETMS Configuration Codes

L2I379

Config ID	Configuration Name	Description
1	SERIAL_NUM	Serial Number
2	HARDWARE_VER	Hardware Version Identifier
3	CPLD_VER	CPLD Version Identifier (Programmable Logic Device)
4	BOOTLOADER_VER	Bootloader Version Identifier
5	FPGA_VER	FPGA Version Identifier (Programmable Logic Device)
6	FIRMWARE_VER	Firmware Version Identifier
7	SOFTWARE_VER	Software Version Identifier
8	COMMON_CFG_FSET_VER	Common Configuration Parameters Fileset Version
9	RR_CFG_FSET_VER	Railroad-Specific Configuration Parameters Fileset Version
10	OPERATION_HRS	Hours of operation
11	KEY_1_PRESS_COUNT	Number of key 1 presses
12	KEY_2_PRESS_COUNT	Number of key 2 presses
13	KEY_3_PRESS_COUNT	Number of key 3 presses
14	KEY_4_PRESS_COUNT	Number of key 4 presses
15	KEY_5_PRESS_COUNT	Number of key 5 presses
16	KEY_6_PRESS_COUNT	Number of key 6 presses
17	KEY_7_PRESS_COUNT	Number of key 7 presses
18	KEY_8_PRESS_COUNT	Number of key 8 presses
19-28	RESERVED	
29	LOCO_ID	Locomotive Identifier
30	GPS_RCVRS	Number of receivers installed on locomotive (1 or 2)
31	GPS1_POS	GPS1 distance from front knuckle
32	GPS2_POS	GPS2 distance from front knuckle
33	GPS1_OFFSET_DIST	GPS1 lateral distance from centerline
34	GPS2_OFFSET_DIST	GPS2 lateral distance from centerline
35	GPS1_OFFSET_DIR	GSP1 offset direction from centerline (left or right)
36	GPS2_OFFSET_DIR	GSP2 offset direction from centerline (left or right)
37	LOCO_WEIGHT	Locomotive weight
38	LOCO_LENGTH	Locomotive length
39	LOCO_HP	Locomotive horsepower
40	RESERVED	
41	LOCO_TRACTION	Locomotive traction type (AC or DC)
42	LOCO_DIAMETER	Maximum locomotive wheel diameter
43	LOCO_AXLE_COUNT	Locomotive axle count
44	LOCO_PPR	Locomotive wheel tach pulses-per-revolution

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Config ID	Configuration Name	Description
45	LOCO_CONTROL_SYS	Locomotive control system interface protocol
46	RESERVED	
47	BUSINESS_CPU	Energy management CPU configuration (present or not)
48	LOCO_WHEEL_CAL_SIZE	Locomotive Calibrated Wheel Size
49	MODEL_NUMBER	Component Model Number
50	CAB_CODE_0	Whether connection is present to receive cab signal code rate 0
51	CAB_CODE_75	Whether connection is present to receive cab signal code rate 75
52	CAB_CODE_120	Whether connection is present to receive cab signal code rate 120
53	CAB_CODE_180	Whether connection is present to receive cab signal code rate 180
54	CAB_CODE_270	Whether connection is present to receive cab signal code rate 270
55	CAB_CODE_75_75	Whether connection is present to receive cab signal code rate 75/75
56	CAB_CODE_120_120	Whether connection is present to receive cab signal code rate 120/120
57	CAB_CODE_180_180	Whether connection is present to receive cab signal code rate 180/180
58	CAB_CODE_270_270	Whether connection is present to receive cab signal code rate 270/270
59	CFG_FILE_PN	Configuration Part Number
60	SW_PART_NUM	Software Part Number
61	CFG_FILE_VER	Configuration File Version
62	WPN	Wabtec Part Number
63	HW_MODE_LEVEL	Hardware Mode Level
64	SOFTWARE_2ND_VER	Secondary Application Version
65	SOFTWARE_CFG_VER	Software Configuration Version
66	KERNEL_VER	Kernel Version
67	OS_VER	O/S Version
68	SYS_PATCH_LEVEL	System Patch Level
69	BIOS_VER	BIOS Version
70	EMB_CONTROL_VER	Embedded Controller Version
71	CPU-10	CPU-10 configuration (present or not)
72	CHM	CHM configuration (present or not)
73	STORAGE_MODEL_NUMBER	Storage Model Number
74	STORAGE_SERIAL_NUMBER	Storage Serial Number
75	STORAGE_FIRMWARE_VERSION	Storage Firmware Version
76	STORAGE_SECTOR_COUNT	Storage 512-byte Sector Count

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Config ID	Configuration Name	Description
77	STORAGE_ERASE_COUNT	Storage Erase Count (average or total)
78	STORAGE_ERASE_LIMIT	Storage Erase Count limit per spec
79	STORAGE_SPARES_LEFT	Storage Spare Blocks Left (%) reported by the card
80	STORAGE_LIFE_LEFT	Storage "life left" percentage reported by the card
81	STORAGE_EST_REM_MONTHS	Storage Estimated Remaining Life (months)
82	PREVIOUS_DAY_NUMBER_YEAR	Number of day of the year, and the year
83	PREVIOUS_DAY_DISTANCE	Distance traveled by the locomotive the previous 24 hour period in feet
84	PREVIOUS_DAY_ACTIVE_DISTANCE	Distance traveled by the locomotive the previous 24 hour period in the ACTIVE state in feet
85	PREVIOUS_DAY_DISENGAGED_DISTANCE	Distance traveled by the locomotive the previous 24 hour period in the DISENGAGED state in feet
86	PREVIOUS_DAY_CUTOUT_DISTANCE	Distance traveled by the locomotive the previous 24 hour period in the CUT OUT state in feet
87	PREVIOUS_DAY_FAILED_DISTANCE	Distance traveled by the locomotive the previous 24 hour period in the FAILED state in feet
88	PREVIOUS_DAY_PTC_ENABLED_DISTANCE	Distance traveled by the locomotive the previous 24 period on PTC enabled track in feet
89	PTC_UTILIZATION	PREVIOUS_DAY_ACTIVE_DISTANCE divided by the PREVIOUS_DAY_PTC_ENABLED_DISTANCE in percentage
90	GPS_CONSTELLATION	Whether the navigation module is configured to use position information from the GPS satellite constellation
91	GALILEO_CONSTELLATION	Whether the navigation module is configured to use position information from the Galileo satellite constellation
92	GLONASS_CONSTELLATION	Whether the navigation module is configured to use position information from the GLONASS satellite constellation
93	BEIDOU_CONSTELLATION	Whether the navigation module is configured to use position information from the BeiDou satellite constellation

If the on-board segment is unable to determine any of the following configuration items for a component, the On-board Component Configuration Report (02088) message will indicate that the configuration item is "UNKNOWN" for that component:

- Software Version Identifier
- Storage Model Number
- Storage Serial Number
- Storage Firmware Version
- Storage 512-byte Sector Count
- Storage Erase Count
- Storage Erase Count limit per spec
- Storage Spare Blocks Left (%) reported by the card
- Storage "life left" percentage reported by the card
- Storage Estimated Remaining Life

If the on-board segment does not have data for a configuration item other than one listed above for a

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component, the On-board Component Configuration Report (02088) message will not report the configuration code. These items will also not show up in the 'Number of Configuration Items' in the (02088) message.

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/// 6 I-ETMS Enhanced Diagnostic Codes

L2I379661 Enhanced Diagnostic Fault Code	Descriptive Text on CDU	Summary of Fault	Brakes Applied	Event Code Mapping
D01 [L2I379661.D 01]	Front BPP sensor inputs are invalid	Invalid Discrete Inputs including the Front BPP	No	0x1D01
D02 [L2I379661.D 02]	TMC Internal Fault	Comm issue between DIO and CPUs.	No	0x1D02
D03 [L2I379661.D 03]	Air Brake Computer is not communicati ng with TMC	Loss of communication with the Electronic Air Brake computer	No	0x1D03
D04 [L2I379661.D 04]	LIG is not communicati ng with TMC	Loss of communication with the LIG	No	0x1D04
D05 [L2I379661.D 05]	Brake Pipe Sensor Inputs are faulty	BPP sensor inputs differ outside of tolerance	No	0x1D05
D06 [L2I379661.D 06]	LIG data is invalid	A data item received from the LIG is invalid	No	0x1D06
D07 [L2I379661.D 07]	Air Brake Computer is not communicati ng with TMC	Loss of communication with the Electronic Air Brake computer	No	0x1D07
D08 [L2I379661.D 08]	Power Cutoff (PCS) Input Invalid	PCS Open Invalid - Usually only occurs in one slice as part of a triplex event	No	0x1D08
D09 [L2I379661.D 09]	Reverser wiring to TMC is faulty	Invalid Discrete Inputs including Reverser handle	No	0x1D09
D10 [L2I379661.D 10]	EBI Horn Circuit Fault	Output from EBI Horn Circuit is invalid	No	0x1D10
D11 [L2I379661.D 11]	Recoverable PTC System Fault	Triplex Mismatch	Yes	0x1D11
D12 [L2I379661.D 12]	Recoverable PTC System Fault	Timing fault receiving an authority	No	0x1D12

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L2I379661 Enhanced Diagnostic Fault Code	Descriptive Text on CDU	Summary of Fault	Brakes Applied	Event Code Mapping
D13 [L2I379661.D 13]	Recoverable PTC System Fault	No braking curves available in ACTIVE	No	0x1D13
D14 [L2I379661.D 14]	Reserved	Reserved	No	N/A
D15 [L2I379661.D 15]	Reserved	Reserved	No	N/A
D16 [L2I379661.D 16]	Reserved	Reserved	No	N/A
D17 [L2I379661.D 17]	Recoverable PTC System Fault	System Recovering	Yes	0x1D17
D18 [L2I379661.D 18]	Recoverable PTC System Fault	Different SW Versions Installed	No	0x1D18
D19 [L2I379661.D 19]	Recoverable PTC System Fault	Out of memory - reboot TMC	No	0x1D19
D20 [L2I379661.D 20]	Recoverable PTC System Fault	CPU Usage Too High	No	0x1D20
D21 [L2I379661.D 21]	Recoverable PTC System Fault	EBI Self test failed	No	0x1D21
D22 [L2I379661.D 22]	TMC Tamper Plate Detected	Tamper Plate has been Removed	No	0x1D22
D23 [L2I379661.D 23]	Reserved	Reserved	No	N/A
D24 [L2I379661.D 24]	TMC Internal Fault	CPU Card not communicating	Yes	0x1D24
D25 [L2I379661.D 25]	Brake Pipe Sensor Inputs are faulty	BPP sensor inputs differ outside of tolerance	No	0x1D25
D26 [L2I379661.D	Air Brake Computer is not	Loss of communication with the	No	0x1D26

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L2I379661 Enhanced Diagnostic Fault Code	Descriptive Text on CDU	Summary of Fault	Brakes Applied	Event Code Mapping
D26	communicating with TMC	Electronic Air Brake computer		
D27 [L2I379661.D 27]	Reserved	Reserved	No	N/A
D28 [L2I379661.D 28]	Recoverable PTC System Fault	Internal Buffer Full	Yes	0x1D28
D29 [L2I379661.D 29]	PTC suspension splits signal block	PTC suspension splits signal block into more than one disjointed section	No	0x1D29
D30 [L2I379661.D 30]	Recoverable PTC System Fault	Multiple CPU faults without automatic reboot when crew is logged off	No	0x1D30
D31 [L2I379661.D 31]	TMC internal fault	IOC not communicating	Yes	0x1D31
D32 [L2I379661.D 32]	Incompatible Operating System	Operating System is not compatible	No	0x1D32
D33 [L2I379661.D 33]	Hardware - Firmware Incompatibility	Incorrect hardware installed or no applicable firmware available.	No	0x1D33
D34 [L2I379661.D 34]	Track configuration is invalid	Invalid combination of Country of Operation and track feature	No	0x1D34

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/// 7 I-ETMS Fault Codes

Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0100 [L2I474.0100]	DIO	DIO: Open Circuit Fault: AI1	Analog input 1 circuit is open	Valid (4-20mA) current in analog input 1 circuit	PTC does not change state	No	For Control System Types ACP, LIG, LSI, NS MC EMU, NS MCEW EMU, QES, TMC-DIO: Analog 1 supplies "BPP1" brake pipe pressure 1. For Control System Types NS 300 EMU, NS HL2 EMU, SR FLRT3 DMU, and SR KSS EMU: Analog 1 supplies "BCP" brake cylinder pressure.	For Control System Types ACP, LIG, LSI, NS MC EMU, NS MCEW EMU, QES, TMC-DIO: Analog 1 supplies "BPP1" brake pipe pressure 1. For Control System Types NS 300 EMU, NS HL2 EMU and SR FLRT3 DMU: DIO Card, BCP sensor, DIO Cable.	Yes	No
0x0101 [L2I474.0101]	DIO	DIO: Open Circuit Fault: AI2	Analog input 2 circuit is open	Valid (4-20mA) current in analog input 2 circuit	PTC does not change state	No	For Control System Types NS MC EMU, NS MCEW EMU and TMC-DIO: Analog 2 supplies "BPP2" brake pipe pressure 2. For Control System Type SR FLRT3 DMU: Analog 2 supplies Tractive Effort. For Control System Types: ACP, LSI, NS HL2 EMU, NS 300 EMU, QES: Analog 2 is not utilized. For Control system type: LIG: Analog 2 supplies "BPP2" brake pipe pressure 2 only when CFG50 is enabled, and therefore this fault will not be raised.	For Control System Types NS MC EMU, NS MCEW EMU, LIG and TMC-DIO: DIO Card, BPP2 sensor, DIO Cable. For Control System Type SR FLRT3 DMU: DIO Card, Tractive Effort Circuit, DIO Cable.	Yes	No
0x0102	DIO	DIO: Open	Analog input	Valid (4-	PTC does	No	For Control System	For Control System	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0102]		Circuit Fault: AI3	3 circuit is open	20mA) current in analog input 3 circuit	not change state		Types NS MC EMU, NS MCEW EMU, TMC-DIO: Analog 3 supplies "BCP" brake cylinder pressure. For Control System Type SR FLRT3 DMU: Analog 3 supplies Brake Effort. For Control System Types: ACP, LIG, LSI, NS HL2 EMU, NS 300 EMU, QES: Analog 3 is not utilized.	Types NS MC EMU, NS MCEW EMU, TMC-DIO: DIO Card, BCP sensor, DIO Cable. For Control System Type SR FLRT3 DMU: DIO Card, Brake Effort Circuit, DIO Cable.		
0x0103 [L2I474.0103]	DIO	DIO: Open Circuit Fault: AI4	Analog input 4 circuit is open	Valid (4-20mA) current in analog input 4 circuit	PTC does not change state	No	For Control System Type TMC-DIO: Analog 4 supplies "ERP" equalizing reservoir pressure. For Control System Types NS MC EMU, NS MCEW EMU, SR FLRT3 DMU: Analog 4 supplies Brake Demand. For Control System Types: ACP, LIG, LSI, NS HL2 EMU, NS 300 EMU, QES: Analog 4 is not utilized. For Control System type SR KSS EMU, Analog 4 supplies Traction/Brake Demand 1	For Control System Type TMC-DIO: DIO Card, ERP sensor, DIO Cable. For Control System Types NS MC EMU, NS MCEW EMU, SR FLRT3 DMU: DIO Card, Brake Demand Circuit, DIO Cable.	Yes	No
0x0104 [L2I474.0104]	DIO	DIO: Open Circuit Fault: AI5	Analog input 5 circuit is open	Valid (4-20mA) current in analog input 5 circuit	PTC does not change state	No	For Control System Types NS 300 EMU, NS HL2 EMU, NS MC EMU, NS MCEW EMU, SR FLRT3 DMU, SR KSS EMU, TMC-DIO: Analog 5 supplies	For Control System Types NS 300 EMU, NS HL2 EMU, NS MC EMU, NS MCEW EMU, SR FLRT3 DMU, TMC-DIO, LIG: DIO Card, ATMOS sensor, DIO	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
							"ATMOS" atmospheric pressure. For Control System Types: ACP, LSI, QES: Analog 5 is not utilized. For Control system type: LIG: Analog 5 supplies "ATMOS" atmospheric pressure only when CFG50 is enabled, and therefore this fault will not be raised.	Cable		
0x0105 [L2I474.0105]	DIO	DIO: Open Circuit Fault: AI6	Analog input 6 circuit is open	Valid (4-20mA) current in analog input 6 circuit	PTC does not change state	No	For Control System Type SR FLRT3 DMU: Analog 6 supplies Traction Demand. For Control System Type ACP, LIG, LSI, NS 300 EMU, NS HL2 EMU, NS MC EMU, NS MCEW EMU, QES, TMC-DIO: Analog 6 is not utilized. For Control System type SR KSS EMU, Analog 6 supplies Traction/Brake Demand 2	For Control System Type SR FLRT3 DMU: DIO Card, Traction Demand Circuit, DIO Cable	Yes	No
0x0106 [L2I474.0106]	DIO	DIO: Short Circuit Fault: AI1	Analog input 1 circuit is shorted	Valid (4-20mA) current in analog input 1 circuit	PTC does not change state	No	For Control System Types ACP, LIG, LSI, NS MC EMU, NS MCEW EMU, QES, TMC-DIO: Analog 1 supplies "BPP1" brake pipe pressure 1. For Control System Types NS 300 EMU, NS HL2 EMU, SR FLRT3 DMU, and SR KSS EMU: Analog 1 supplies "BCP" brake cylinder	For Control System Types ACP, LIG, LSI, NS MC EMU, NS MCEW EMU, QES, TMC-DIO: DIO Card, BPP1 sensor, DIO Cable. For Control System Types NS 300 EMU, NS HL2 EMU and SR FLRT3 DMU: DIO Card, BCP sensor, DIO	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
							pressure.	Cable.		
0x0107 [L2I474.0107]	DIO	DIO: Short Circuit Fault: AI2	Analog input 2 circuit is shorted	Valid (4-20mA) current in analog input 2 circuit	PTC does not change state	No	For Control System Types NS MC EMU, NS MCEW EMU and TMC-DIO: Analog 2 supplies "BPP2" brake pipe pressure 2. For Control System Type SR FLRT3 DMU: Analog 2 supplies Tractive Effort. For Control System Types: ACP, LSI, NS HL2 EMU, NS 300 EMU, QES: Analog 2 is not utilized. For Control system type: LIG: Analog 2 supplies "BPP2" brake pipe pressure 2 only when CFG50 is enabled, and therefore this fault will not be raised.	For Control System Types NS MC EMU, NS MCEW EMU, LIG and TMC-DIO: DIO Card, BPP2 sensor, DIO Cable. For Control System Type SR FLRT3 DMU: DIO Card, Tractive Effort Circuit, DIO Cable.	Yes	No
0x0108 [L2I474.0108]	DIO	DIO: Short Circuit Fault: AI3	Analog input 3 circuit is shorted	Valid (4-20mA) current in analog input 3 circuit	PTC does not change state	No	For Control System Types NS MC EMU, NS MCEW EMU, TMC-DIO: Analog 3 supplies "BCP" brake cylinder pressure. For Control System Type SR FLRT3 DMU: Analog 3 supplies Brake Effort. For Control System Types: ACP, LIG, LSI, NS HL2 EMU, NS 300 EMU, QES: Analog 3 is not utilized.	For Control System Types NS MC EMU, NS MCEW EMU, TMC-DIO: DIO Card, BCP sensor, DIO Cable. For Control System Type SR FLRT3 DMU: DIO Card, Brake Effort Circuit, DIO Cable.	Yes	No
0x0109	DIO	DIO: Short	Analog input	Valid (4-	PTC does	No	For Control System	For Control System	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0109]		Circuit Fault: AI4	4 circuit is shorted	20mA) current in analog input 4 circuit	not change state		Type TMC-DIO: Analog 4 supplies "ERP" equalizing reservoir pressure. For Control System Types NS MC EMU, NS MCEW EMU, SR FLRT3 DMU: Analog 4 supplies Brake Demand. For Control System Types: ACP, LIG, LSI, NS HL2 EMU, NS 300 EMU, QES: Analog 4 is not utilized. For Control System type SR KSS EMU, Analog 4 supplies Traction/Brake Demand 1	Type TMC-DIO: DIO Card, ERP sensor, DIO Cable. For Control System Types NS MC EMU, NS MCEW EMU, SR FLRT3 DMU: DIO Card, Brake Demand Circuit, DIO Cable.		
0x010A [L2I474.010A]	DIO	DIO: Short Circuit Fault: AI5	Analog input 5 circuit is shorted	Valid (4-20mA) current in analog input 5 circuit	PTC does not change state	No	For Control System Types NS 300 EMU, NS HL2 EMU, NS MC EMU, NS MCEW EMU, SR FLRT3 DMU, SR KSS EMU, TMC-DIO: Analog 5 supplies "ATMOS" atmospheric pressure. For Control System Types: ACP, LSI, QES: Analog 5 is not utilized. For Control system type: LIG: Analog 5 supplies "ATMOS" atmospheric pressure only when CFG50 is enabled, and therefore this fault will not be raised.	For Control System Types NS 300 EMU, NS HL2 EMU, NS MC EMU, NS MCEW EMU, SR FLRT3 DMU, TMC-DIO, LIG: DIO Card, ATMOS sensor, DIO Cable	Yes	No
0x010B [L2I474.010B]	DIO	DIO: Short Circuit Fault: AI6	Analog input 6 circuit is shorted	Valid (4-20mA) current in analog input	PTC does not change state	No	For Control System Type SR FLRT3 DMU: Analog 6 supplies	For Control System Type SR FLRT3 DMU: DIO Card, Traction Demand Circuit, DIO	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
				6 circuit			Traction Demand. For Control System Type ACP, LIG, LSI, NS 300 EMU, NS HL2 EMU, NS MC EMU, NS MCEW EMU, QES, TMC-DIO: Analog 6 is not utilized. For Control System type SR KSS EMU, Analog 6 supplies Traction/Brake Demand 2	Cable		
0x010C [L2I474.010C]	DIO	DIO: Analog Out of Range: AI7	Analog input is outside of 0 to 75 volt range	Voltage input needs to be between 0 to 75 volts	Degraded Energy Management Functionality	No		DIO Card	Yes	No
0x010D [L2I474.010D]	DIO	DIO: Card Fault	Analog not calibrated during manufacturing process or flash has corrupt data	Replace DIO Card	PTC does not change state	No		DIO Card	Yes	No
0x010E [L2I474.010E]	DIO	DIO: Card Fault	Analog not calibrated during manufacturing process or flash has corrupt data	Replace DIO Card	PTC does not change state	No		DIO Card	Yes	No
0x010F [L2I474.010F]	DIO	DIO: Card Fault	Analog not calibrated during manufacturing process or flash has corrupt data	Replace DIO Card	PTC does not change state	No		DIO Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0110 [L2I474.0110]	DIO	DIO: Card Fault	Analog not calibrated during manufacturing process or flash has corrupt data	Replace DIO Card	PTC does not change state	No		DIO Card	Yes	No
0x0111 [L2I474.0111]	DIO	DIO: Card Fault	Analog not calibrated during manufacturing process or flash has corrupt data	Replace DIO Card	PTC does not change state	No		DIO Card	Yes	No
0x0112 [L2I474.0112]	DIO	DIO: Card Fault	Analog not calibrated during manufacturing process or flash has corrupt data	Replace DIO Card	PTC does not change state	No		DIO Card	Yes	No
0x0113 [L2I474.0113]	DIO	DIO: Card Fault	Analog not calibrated during manufacturing process or flash has corrupt data	Replace DIO Card	PTC does not change state	No		DIO Card	Yes	No
0x0114 [L2I474.0114]	DIO	DIO: Generator Voltage Modification missing	Version of DIO card does not support generator voltage input	DIO Card supports generator voltage input	Degraded Energy Management Functionality	No		DIO Card	Yes	No
0x0115 [L2I474.0115]	PTC	DIO: No Acceptable Firmware Installed	Hardware type is identified, but no acceptable firmware	Does not clear	System State goes to FAILED	No		DIO module	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			version is installed							
0x0116 [L2I474.0116]	PTC	IOC: No Acceptable Firmware Installed	Hardware type is identified, but no acceptable firmware version is installed	Does not clear	System State goes to FAILED	No		IOC module	Yes	No
0x0117 [L2I474.0117]	PTC	RSM: No Acceptable Firmware Installed	Hardware type is identified, but no acceptable firmware version is installed	Does not clear	System State goes to FAILED	No		RSM module	Yes	No
0x0150 [L2I474.0150]	EBI	EBI: Card Fault	32 volt power supply on EBI card not functional	32 volt power supply is functional	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0151 [L2I474.0151]	EBI	EBI: Card Fault	Self test of EBI drive circuit failed	NA - Fault will remain active	PTC Slice State goes to FAILED	No		Power cycle power on TMC, EBI Card	Yes	No
0x0152 [L2I474.0152]	EBI	EBI: Card Fault	U1 Micro-controller fault on EBI board	Micro-controller clears internal fault	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0153 [L2I474.0153]	EBI	EBI: Card Fault	U2 Micro-controller fault on EBI board	Micro-controller clears internal fault	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0154 [L2I474.0154]	EBI	EBI: Card Fault	U3 Micro-controller fault on EBI board	Micro-controller clears internal fault	PTC Slice State goes to FAILED	No		EBI Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0155 [L2I474.0155]	EBI	EBI: Card Fault	Internal EBI card fault reported by EBI	No internal EBI card faults reported by EBI	PTC does not change state	No		EBI Card	Yes	No
0x0156 [L2I474.0156]	EBI	EBI: Card Fault	External EBI card fault reported by EBI	No external EBI card faults reported by EBI	PTC does not change state	No		EBI Card	Yes	No
0x0157 [L2I474.0157]	EBI	EBI: Card Fault	EBI HW revision does not match I-ETMS system configuration	EBI HW revision matches I-ETMS system configuration	PTC Slice State goes to FAILED	Yes		EBI Card	Yes	No
0x0158 [L2I474.0158]	EBI2	EBI2: Card Fault	EBI2 HW revision does not match I-ETMS system configuration: EBI-301	EBI HW revision matches I-ETMS system configuration: EBI-PCR	PTC State goes to FAILED	Yes	Applies to Control System Type: SR KSS EMU	EBI2 Card	Yes	No
0x0159 [L2I474.0159]	EBI2	EBI2: Signal Fault	At least one EBI2 pull up signal is set LOW(0): • PCR Pull Up1 • PCR Pull Up2	Both EBI2 pull up signals are set HIGH(1): • PCR Pull Up1 • PCR Pull Up2	PTC State goes to FAILED	Yes	Applies to Control System Type: SR KSS EMU	EBI2 Card	Yes	No
0x0160 [L2I474.0160]	EBI2	EBI2: Card Fault	32 volt power supply on	32 volt power supply is functional	PTC State goes to FAILED	No	Applies to Control System Type: SR KSS EMU	EBI2 Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			EBI2 card not functional							
0x0161 [L2I474.0161]	EBI2	EBI2: Card Fault	Self test of EBI2 drive circuit failed	NA - Fault will remain active	PTC State goes to FAILED	No	Applies to Control System Type: SR KSS EMU	Power cycle power on TMC, EBI2 Card	Yes	No
0x0162 [L2I474.0162]	EBI2	EBI2: Card Fault	U1 Micro-controller fault on EBI board	Micro-controller clears internal fault	PTC State goes to FAILED	No	Applies to Control System Type: SR KSS EMU	EBI2 Card	Yes	No
0x0163 [L2I474.0163]	EBI2	EBI2: Card Fault	U2 Micro-controller fault on EBI board	Micro-controller clears internal fault	PTC State goes to FAILED	No	Applies to Control System Type: SR KSS EMU	EBI2 Card	Yes	No
0x0164 [L2I474.0164]	EBI2	EBI2: Card Fault	U3 Micro-controller fault on EBI board	Micro-controller clears internal fault	PTC State goes to FAILED	No	Applies to Control System Type: SR KSS EMU	EBI2 Card	Yes	No
0x0200 [L2I474.0200]	IOC	IOC: Card Fault	Communication problem between FPGA and Processor on IOC	FPGA is functional and communicating with Processor	PTC Slice State goes to FAILED	No		IOC Card	Yes	No
0x0201 [L2I474.0201]	IOC	IOC: Card Fault	Communication problem between FPGA and Processor on IOC	FPGA is functional and communicating with Processor	PTC Slice State goes to FAILED	No		IOC Card	Yes	No
0x0202 [L2I474.0202]	IOC	IOC: Card Fault	Problem with on-board IOC flash	Flash memory is operating correctly	PTC Slice State goes to FAILED	No		IOC Card	Yes	No
0x0210 [L2I474.0210]	DIO	DIO: Card Fault	Communication problem between FPGA and	FPGA is functional and communicating	PTC Slice State goes to FAILED	No		DIO Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			Processor on DIO	ng with Processor						
0x0211 [L2I474.0211]	DIO	DIO: Card Fault	Communication problem between FPGA and Processor on DIO	FPGA is functional and communicating with Processor	PTC Slice State goes to FAILED	No		DIO Card	Yes	No
0x0300 [L2I474.0300]	DIO	DIO: Card Fault	Problem with on-board DIO flash	Replace the DIO card.	PTC does not change state	No		DIO Card	Yes	No
0x0301 [L2I474.0301]	DIO	DIO: Card Fault	Communication problem between DIO and CPUs	Communication restored between DIO and CPUs	PTC does not change state	No		DIO Card	Yes	No
0x0302 [L2I474.0302]	DIO	DIO: Card Fault	Unable to write to DIO Flash	Replace the DIO card.	PTC does not change state	No		DIO Card	Yes	No
0x0303 [L2I474.0303]	DIO	DIO: Comm Fault	Communication problem between DIO and CPUs	Communication restored between DIO and CPUs	PTC does not change state	No		DIO Card	Yes	No
0x0304 [L2I474.0304]	DIO	DIO: Comm Fault	Communication problem between DIO and CPUs	Communication restored between DIO and CPUs	PTC does not change state	No		DIO Card	Yes	No
0x0310 [L2I474.0310]	IOC	IOC: Card Fault	Problem with on-board IOC flash	Replace the IOC card	PTC does not change state	No		IOC Card	Yes	No
0x0311 [L2I474.0311]	IOC	IOC: Card Fault	Communication problem between IOC and	Communication restored between IOC and CPUs	PTC does not change state	No		IOC Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			CPUs							
0x0312 [L2I474.0312]	IOC	IOC: Card Fault	Unable to write to r IOC Flash	Replace the IOC card.	PTC does not change state	No		IOC Card	Yes	No
0x0313 [L2I474.0313]	IOC	IOC: Comm Fault	Communication problem between IOC and CPUs	Communication restored between IOC and CPUs	PTC does not change state	No		IOC Card	Yes	No
0x0314 [L2I474.0314]	IOC	IOC: Comm Fault	Communication problem between IOC and CPUs	Communication restored between IOC and CPUs	PTC does not change state	No		IOC Card	Yes	No
0x0333 [L2I474.0333]	EOT	EOT: Comm Fault	Sequence number of received packet is not what was expected (must increment by 1)	Receive an HOT packet with the expected sequence number	PTC does not change state but will use more conservative brake algorithm if in TMC-DIO configuration	No		CPU, DIO, or HOT	Yes	No
0x0334 [L2I474.0334]	EOT	EOT: Comm Fault	Checksum of received HOT message did not validate for 3 consecutive sets of 3 messages	Receive an HOT message with a good checksum	PTC does not change state but will use more conservative brake algorithm if in TMC-DIO configuration	No		CPU, DIO, or HOT	Yes	No
0x0335 [L2I474.0335]	EOT	EOT Comm Loss	EOT Pair Status is Paired (PTC EOT) and EOT Comm State is Not	EOT Pair Status is not Paired (PTC EOT) or EOT Comm State is	PTC does not change state	No		HOTCC, EOT Service or EOT	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			Communicating	Communicating						
0x0336 [L2I474.0336]	EOT	EOT Pair Failed	EOT Pair Status is Pair Failed	EOT Pair Status is NOT Pair Failed	PTC does not change state	No		HOTCC, EOT Service	Yes	No
0x0337 [L2I474.0337]	EOT	EOT Paired to Other Locomotive	EOT Pair Status is Paired to Other Locomotive	EOT Pair Status is NOT Paired to Other Locomotive	PTC does not change state	No			Yes	No
0x0338 [L2I474.0338]	HOTCC	HOTCC: Class D Comm Loss	HOTCC interface enabled and Onboard is unable to establish Class D link with HOTCC	Onboard restored Class D link with HOTCC	PTC does not change state but will not be able to communicate with HOTCC	No		CPU Card	Yes	No
0x0339 [L2I474.0339]	HOTCC	HOTCC: Class D Buffer Full	HOTCC Class D queue overflows	HOTCC Class D queue has room for a new message	PTC does not change state, any new message to HOTCC is dropped	No		CPU Code	Yes	No
0x0340 [L2I474.0340]	EOT	EOT GPS Loss	EOT Pair Status has been Paired (PTC EOT) for at least 600 seconds and EOT Location (2262) message has not been	EOT Pair Status is not Paired (PTC EOT) or EOT Location (2262) message has been received within the last 600 seconds	PTC does not change state	No		HOTCC or EOT	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			received within the last 600 seconds							
0x0341 [L2I474.0341]	EOT	EOT Comm Loss	HOT Link Status is HOT Linked and HOT Comm State is not communicating	HOT Link Status is Not Linked or HOT Comm State is communicating	PTC does not change state	No		HOTCC, HOT or EOT	Yes	No
0x0342 [L2I474.0342]	EOT	EOT Camera Comm Loss	EOT Pair Status is Paired (PTC EOT) and EOT Camera State has been Slow Rate or Fast Rate for at least 180 seconds and EOT Image (2263) message has not been received within the last 180 seconds	EOT Pair Status is not Paired (PTC EOT) or EOT Camera State is Off or EOT Image (2263) message has been received within the last 180 seconds	PTC does not change state	No		HOTCC or EOT	Yes	No
0x0343 [L2I474.0343]	EOT	EOT GPS Loss	HOT Link Status has been HOT Linked for at least 600 seconds	HOT Link Status is Not Linked or EOT Location (2262)	PTC does not change state	No		HOTCC, HOT, or EOT	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			and EOT Location (2262) message has not been received within the last 600 seconds	message has been received within the last 600 seconds						
0x0505 [L2I474.0505]	EBI2	EBI2:Phase Break is Cutout or Open	Phase Break is cut out or open	Phase Break is cut in	PTC does not change state	No	Control System Type: SR KSS EMU	Phase Break cable	Yes	No
0x0506 [L2I474.0506]	EBI	EBI: Vital Emergency Brakes are Cutout or Open	PTC Vital Emergency brakes cut out or open seen on vital emergency brake circuit	PTC vital emergency brake cutout switch is in cut in position and no open circuit	PTC State Goes to CUT OUT	No	Control System Type SR FLRT3 DMU SR KSS EMU Application Fault associated with Cut Out Assembly (COA) setting. If COA Emergency switch is set to Cut Out, then 0x0506 fault code is raised for a slot6 EBI301 configuration.	EBI Cable	Yes	Yes
0x0507 [L2I474.0507]	EBI2	EBI2: Card Fault	Phase break command has been sent to active, but the EBI-A Discrete remains set to:	Phase break command has been sent to active and the EBI- A Discrete is set to: H-EMG- OUTPUT =	PTC does not change state	No	Control System Type SR KSS EMU Application Fault associated with the phase break command	EBI Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			H-EMG-OUTPUT = (0) for > 3 seconds after phase break command was set to active	(1) for > 3 seconds after phase break command was set to active			feedback/status from the slot 9 EBI.			
0x0508 [L2I474.0508]	EBI	EBI: Horn mag valve short	Short on horn mag valve circuit	Valid current on horn circuit	PTC displays "HORN" flag, does not change state	No		EBI Card or EBI Cable	Yes	No
0x0509 [L2I474.0509]	EBI	EBI: Penalty brake open	Open seen on penalty brake circuit	No open circuit on penalty brake circuit	PTC does not change state	No		EBI Card or EBI Cable	Yes	No
0x050A [L2I474.050A]	EBI	EBI: Emergency brake short	Short on emergency circuit	Valid current on emergency circuit	PTC state does not change	No		EBI Card or EBI Cable	Yes	No
0x050B [L2I474.050B]	EBI	EBI: Penalty brake short	Short on penalty circuit	Valid current on penalty circuit	PTC Slice State goes to FAILED	Yes		EBI Card or EBI Cable	Yes	No
0x050C [L2I474.050C]	TMC	TMC: EBI Comm Fault	EBI communication problem in TMC	EBI Communication is restored in TMC	PTC Slice State goes to FAILED	Yes		EBI or CPU Card	Yes	No
0x050D [L2I474.050D]	TMC	TMC: EBI Comm Fault	EBI Communication problem in TMC	EBI Communication is restored in TMC	PTC does not change state	No		EBI or CPU Card	Yes	No
0x050E [L2I474.050E]	EBI	EBI: Card Fault	EBI Self test failed	NA - Fault will remain active	PTC Slice State goes to FAILED	No		Power cycle power on TMC, EBI Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x050F [L2I474.050F]	EBI	EBI: Penalty Brakes are Cutout or Open	PTC Penalty brake switch is in cutout position	PTC penalty cutout switch in cut in position and no open circuit	PTC Slice State goes to CUT OUT	No	Control System Type NS MC EMU NS MCEW EMU NS 300 EMU NS HL2 EMU TMC-DIO LIG LSI ACP HR EMU Application Fault associated with Cut Out Assembly (COA) setting. If Penalty Brake is set to Cut Out on the COA, then 0x050F fault code is raised.	EBI Cable	Yes	Yes
0x0510 [L2I474.0510]	EBI	EBI: Emergency Brakes are Cutout or Open	PTC Emergency brakes cut out or open seen on emergency brake circuit	PTC emergency brake cutout switch is in cut in position and no open circuit	PTC does not include emergency capability in the braking algorithm	No	Control System Type NS MC EMU NS MCEW EMU NS 300 EMU NS HL2 EMU TMC-DIO LIG LSI ACP HR EMU	EBI Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
							Application Fault associated with Cut Out Assembly (COA) setting. If Emergency Brake is set to Cut Out on the COA, then 0x0510 fault code is raised.			
0x0511 [L2I474.0511]	EBI	EBI: Horn is Cutout or Open	PTC Horn cut out or open seen on horn circuit	PTC horn cutout switch is in cut in position and no open circuit	PTC loses capability to blow horn, displays "HORN" flag, does not change state	No		EBI Cable	Yes	No
0x0512 [L2I474.0512]	CPUx	CPUx: Software Fault	Comm port on the TMC is open	Close communication port on TMC	PTC Slice State goes to FAILED	No		EBI, CPU Card	Yes	No
0x0513 [L2I474.0513]	TMC	TMC: EBI Comm Fault	EBI communication problem in TMC	EBI Communication is restored in TMC	PTC does not change state	No		EBI or CPU Card	Yes	No
0x0514 [L2I474.0514]	CPUx	CPUx: Software Fault	Software object problem	Software object receives new command or software object functionality is restored	PTC Slice State goes to FAILED	Yes		Onboard software on CPU or CPU Card	Yes	No
0x0515 [L2I474.0515]	CPUx	CPUx: Software Fault	Software object problem	Software object receives new command or software	PTC Slice State goes to FAILED	Yes		EBI Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
				object functionality is restored						
0x0516 [L2I474.0516]	TMC	TMC: EBI Comm Fault	EBI communication problem in TMC	EBI Communication is restored in TMC	PTC Slice State goes to FAILED	Yes		EBI or CPU Card	Yes	No
0x0517 [L2I474.0517]	TMC	TMC: Simplex Strap Fault	PTC application detects EBI strap for simplex	PTC application detects EBI is configured for triplex	PTC Slice State goes to FAILED	No		TMC	Yes	No
0x0518 [L2I474.0518]	EBI	EBI: Non-Vital Penalty Brakes are Cutout or Open	PTC Non-Vital Penalty brake switch is in cutout position	PTC non-vital penalty cutout switch in cut in position and no open circuit	PTC State goes to CUT OUT	No	Control System Type SR FLRT3 DMU SR KSS EMU Application Fault associated with Cut Out Assembly (COA) setting. If Non-Vital Penalty Brake is set to Cut Out on the COA, then 0x0518 fault code is raised for a slot6 EBI301 configuration.	EBI Cable	Yes	No
0x0519 [L2I474.0519]	EBI	EBI: Horn Button Stuck or Shorted	Horn button stuck on or shorted	Horn button open or valid current on horn circuit	horn blows continuously, displays "HORN" flag, does not change state	No		EBI Card	Yes	No
0x051A [L2I474.051A]	TMC	TMC: DIO Comm Fault	Loss of comm between	Communication restored between	PTC Slice State goes	Yes		DIO or CPU Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			CPU and DIO	CPU and DIO	to FAILED					
0x051B [L2I474.051B]	TMC	TMC: DIO Comm Fault	Loss of comm between CPU and DIO	Communication restored between CPU and DIO	PTC Slice State goes to FAILED	Yes		DIO or CPU Card	Yes	No
0x051C [L2I474.051C]	GPS1	GPS1: Comm Fault	Onboard fail to open a UDP listening socket for GPS 1	Onboard opens UDP listening socket for GPS1	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x051D [L2I474.051D]	GPS2	GPS2: Comm Fault	Onboard fail to open a UDP listening socket for GPS 2	Onboard opens UDP listening socket for GPS2	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x051E [L2I474.051E]	GPS1	GPS1: Comm Fault	Onboard fail to initialize lib_serial_io for GPS 1	Onboard initializes lib_serial_io for GPS 1	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x051F [L2I474.051F]	GPS2	GPS2: Comm Fault	Onboard fail to initialize lib_serial_io for GPS 2	Onboard initializes lib_serial_io for GPS 2	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0520 [L2I474.0520]	GPS1	GPS1: Loss of Signal	GPS 1 sent message that has been declared low quality	GPS 1 sends a high quality message	No state change if only one GPS comm link is down, otherwise DeadReckon until track uncertainty reached its max limit	No		GPS, IOC Cable, or IOC Card	Yes	No
0x0521	GPS2	GPS2: Loss	GPS 2 sent	GPS 2 sends	No state	No		GPS, IOC Cable, or	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0521]		of Signal	message that has been declared low quality	a high quality message	change if only one GPS comm link is down, otherwise DeadReckon until track uncertainty reached its max limit			IOC Card		
0x0522 [L2I474.0522]	GPS1	GPS1: Invalid Data	Onboard receives a bad GGA/RMC packet from GPS 1	Onboard receives a valid GGA/RMC packet from GPS 1	No state change if only one GPS has invalid data, otherwise DeadReckon until track uncertainty reached its max limit	No		GPS, IOC Cable, or IOC Card	Yes	No
0x0523 [L2I474.0523]	GPS2	GPS2: Invalid Data	Onboard receives a bad GGA/RMC packet from GPS 2	Onboard receives a valid GGA/RMC packet from GPS 2	No state change if only one GPS has invalid data, otherwise DeadReckon until track uncertainty reached its max limit	No		GPS, IOC Cable, or IOC Card	Yes	No
0x0524 [L2I474.0524]	GPS1	GPS1: Comm Lost	No GPS messages received from GPS 1 for 3 seconds	Valid GPS message received from GPS 1	No state change if only one GPS comm link is down, otherwise DeadReckon until track uncertainty reached its	No		GPS, IOC Cable, or IOC Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
					max limit					
0x0525 [L2I474.0525]	GPS2	GPS2: Comm Lost	No GPS messages received from GPS 2 for 3 seconds	Valid GPS message received from GPS 2	No state change if only one GPS comm link is down, otherwise DeadReckon until track uncertainty reached its max limit	No		GPS, IOC Cable, or IOC Card	Yes	No
0x0526 [L2I474.0526]	GPS1	GPS1: No GGA Data	No GPS-GGA messages received from GPS 1 for 3 seconds	Valid GPS-GGA message received from GPS 1	No state change if only one GPS is missing GGA data, otherwise DeadReckon until track uncertainty reached its max limit	No		GPS, IOC Cable, or IOC Card	Yes	No
0x0527 [L2I474.0527]	GPS2	GPS2: No GGA Data	No GPS-GGA messages received from GPS 2 for 3 seconds	Valid GPS-GGA message received from GPS 2	No state change if only one GPS is missing GGA data, otherwise DeadReckon until track uncertainty reached its max limit	No		GPS, IOC Cable, or IOC Card	Yes	No
0x0528 [L2I474.0528]	GPS1	GPS1: No RMC Data	No GPS-RMC messages received from GPS 1 for 3	Valid GPS-RMC message received from GPS 1	No state change if only one GPS is missing RMC data,	No		GPS, IOC Cable, or IOC Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			seconds		otherwise DeadReckon until track uncertainty reached its max limit					
0x0529 [L2I474.0529]	GPS2	GPS2: No RMC Data	No GPS-RMC messages received from GPS 2 for 3 seconds	Valid GPS-RMC message received from GPS 2	No state change if only one GPS is missing RMC data, otherwise DeadReckon until track uncertainty reached its max limit	No		GPS, IOC Cable, or IOC Card	Yes	No
0x052C [L2I474.052C]	Tamper Switch	Tamper Switch: Intrusion Detected	Tamper plate has been removed for > 5 sec	Install tamper plate	PTC Slice State goes to FAILED	No		Tamper Bar or TMC	Yes	No
0x052D [L2I474.052D]	TMC	TMC: IOC Comm Fault	No comm from IOC	Communication restored on IOC	PTC Slice State goes to FAILED	Yes		IOC Card	Yes	No
0x052E [L2I474.052E]	LocoID	LocoID: Module Fault	CPU/IOC unable to communicate with LocoID	Communication restored between CPU and IOC/LocoID	PTC does not change state	No		IOC Card, CPU Card, or LocoID Cable	Yes	No
0x052F [L2I474.052F]	LocoID	LocoID: Module Fault	CPU/IOC unable to communicate with LocoID	Communication restored between CPU and IOC/LocoID	PTC does not change state	No		IOC Card, CPU Card, or LocoID Cable	Yes	No
0x0530 [L2I474.0530]	LocoID	LocoID: Module Fault	Communication problem between IOC and	Communication restored between IOC and Loco ID	PTC does not change state	No		Loco ID EEPROM, IOC Cable, or IOC Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			Loco ID EEPROM, or problem with Loco ID EEPROM	EEPROM						
0x0531 [L2I474.0531]	LocoID	LocoID: Loco Id missing	Unable to read Loco ID	Able to read Loco ID	PTC does not change state	No		Loco ID EEPROM, IOC Cable, or IOC Card	Yes	No
0x0532 [L2I474.0532]	CPUx	CPUx: Loco ID Comm Fault	CPU/IOC unable to communicate with LocoID	Communication restored between IOC and Loco ID EEPROM	PTC does not change state	No		Loco ID EEPROM, IOC Cable, or IOC Card	Yes	No
0x0533 [L2I474.0533]	TMC	TMC: IOC Comm Fault	No comm from IOC	Communication restored from IOC	PTC does not change state	No		IOC Cable, or IOC Card	Yes	No
0x0535 [L2I474.0535]	LocoID	LocoID: Module Fault	LocoID EEPROM data read but invalid version	Valid version of LocoID EEPROM	PTC does not change state	No		Loco ID EEPROM	Yes	No
0x0536 [L2I474.0536]	LocoID	LocoID: Module Fault	LocoID EEPROM data read but invalid version	Valid version of LocoID EEPROM	PTC does not change state	No		Loco ID EEPROM	Yes	No
0x0537 [L2I474.0537]	LocoID	LocoID: Module Fault	LocoID EEPROM data read but invalid version	Valid version of LocoID EEPROM	PTC does not change state	No		Loco ID EEPROM	Yes	No
0x0538 [L2I474.0538]	LocoID	LocoID: Module Fault	LocoID EEPROM data read but invalid version	Valid version of LocoID EEPROM	PTC does not change state	No		Loco ID EEPROM	Yes	No
0x0539	LocoID	LocoID: Module Fault	LocoID EEPROM data read	Valid version of LocoID	PTC does not change	No		Loco ID EEPROM	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0539]			but invalid version	EEPROM	state					
0x053B [L2I474.053B]	LocoID	LocoID: Module Fault	LocoID EEPROM data read but invalid version	Valid version of LocoID EEPROM	PTC does not change state	No		Loco ID EEPROM	Yes	No
0x053C [L2I474.053C]	LocoID	LocoID: Module Fault	LocoID EEPROM data read but invalid version	Valid version of LocoID EEPROM	PTC does not change state	No		Loco ID EEPROM	Yes	No
0x053D [L2I474.053D]	LocoID	LocoID: Configuration Fault	Configuration issue with the Loco ID data	Configuration issue resolved with the Loco ID data	PTC Slice State goes to FAILED	No		Loco ID EEPROM	Yes	No
0x053E [L2I474.053E]	LocoID	LocoID: LocoID Unlocked	Loco ID unlocked	LocoID locked	PTC does not change state	No		LocoID EEPROM	Yes	No
0x053F [L2I474.053F]	LocoID	LocoID: Module Fault	LocoID EEPROM data read but invalid version	Valid version of LocoID EEPROM	PTC does not change state	No		LocoID EEPROM	Yes	No
0x0540 [L2I474.0540]	CSM	CSM: IOC Comm Fault	Error opening or closing serial port to the IOC	Open or close serial port to IOC	PTC does not change state	No		IOC Cable, or IOC Card	Yes	No
0x0541 [L2I474.0541]	TMC	TMC: IOC Comm Fault	No comm from IOC	Communication restored from IOC	Does not change state. Cab signal treated as cutout.	No		IOC Cable, or IOC Card	Yes	No
0x0551	AG	AG: Class D Comm Loss	Onboard is not able to establish	Onboard restored Class D link	PTC does not change state, but	No		Review params_comm_RR.cfg	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0551]			Class D link with AG	with AG	will be unable to use off board comm			file		
0x0553 [L2I474.0553]	CHM	CHM: Class D Comm Loss	Onboard is not able to establish Class D link with Event Recorder	Onboard restored Class D link with Event Recorder	PTC does not change state, yellow recorder flag is displayed	No		Review params_comm_RR.cfg file	Yes	No
0x0554 [L2I474.0554]	EM	EM: Class D Comm Loss	Onboard not able to establish class d link between primary CPU and CPU4	Onboard restored Class D link with EM	PTC does not change state, but will loose EM capability	No		Review params_comm_RR.cfg file, or EM application	Yes	No
0x0555 [L2I474.0555]	CPUx	CPUx: Class D Comm Loss	Onboard not able to establish class D link	Onboard restored Class D link	PTC does not change state, but will loose EM capability	No		Review params_comm_RR.cfg file, or EM application	Yes	No
0x0556 [L2I474.0556]	AG	AG: Class D Buffer Full	AG Class D queue overflows	AG Class D buffer queue has room for a new message	PTC does not change state, any new message to AG is dropped	No		CPU Card	Yes	No
0x0558 [L2I474.0558]	CHM	CHM: Class D Buffer Full	ER Class D queue overflows	ER Class D queue has room for a new message	Any new message to CHM is dropped, yellow recorder flag is displayed, Queue is	No		CPU Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
					flushed when connection reestablished.					
0x0559 [L2I474.0559]	EM	EM: Class D Buffer Full	EM Class D queue overflows	EM Class D queue has room for a new message	PTC does not change state, any new message to EM is dropped	No		CPU Card	Yes	No
0x0560 [L2I474.0560]	CDUx	CDUx: Sonalert Fault	I2C comm error	NA - Unrecoverable Fault	PTC does not change state	No		CDU	Yes	No
0x0561 [L2I474.0561]	CHM	EVR: Class D Comm	Onboard is not able to establish Class D link with Event Recorder	Onboard restored Class D link with Event Recorder	CDU Screen will not be recorded	No		Review params_comm_RR.cfg file	Yes	No
0x0562 [L2I474.0562]	CHM	EVR: Maintenance Indicator	EVR indicates CHM Maintenance required	EVR indicates no CHM Maintenance is required or ER Status (2003) message has not been received for data_ack_timeout seconds	PTC does not change state.	No		CHM	Yes	No
0x0563 [L2I474.0563]	CHM	EVR: SDD Maintenance Indicator	EVR indicates Removable SDD Maintenance required	EVR indicates no Removable SSD Maintenance is required or ER Status (2003)	PTC does not change state.	No		CHM	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
				message has not been received for data_ack_timeout seconds						
0x0564 [L2I474.0564]	CHM	CHM: Event Data Not Recorded	CHM indicates Event Data Not Recorded	CHM indicates Event Data is recorded or CHM Status (4500) message has not been received for data_ack_timeout seconds	PTC does not change state, Yellow recorder flag is displayed	No		CHM	Yes	No
0x0565 [L2I474.0565]	CHM	EVR: Authentication Failed	EVR failed the Challenge/Authentication check	EVR passes the Challenge/Authentication check	PTC does not change state.	No		CHM	Yes	No
0x0566 [L2I474.0566]	CHM	EVR: CDU Image Not Recorded	EVR indicates CDU Image not Recorded	EVR indicates CDU Image is recorded or ER Status (2003) message has not been received for data_ack_timeout seconds	PTC does not change state.	No		CHM	Yes	No
0x0567 [L2I474.0567]	CHM	EVR: Stream Buffer Full	Stream Buffer has overflowed	Stream Buffer has room for a new data	CDU Screen will not be recorded. Queue will be flushed when connection reestablished	No		CPU Card	Yes	No

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					d.					
0x05A0 [L2I474.05A0]	EAB	EAB: Connection Failure	Unable to open or initialize the hardware connection to the EAB (via the IOC).	NA - Unrecoverable Fault	PTC Slice State goes to FAILED	No		TMC	Yes	No
0x05A1 [L2I474.05A1]	EAB	EAB: Comm Loss	Loss of comm from the EAB.	Communication restored with EAB	PTC does not change state	No		EAB Cable, IOC Card, Loco ID Control System Configuration	Yes	No
0x05A2 [L2I474.05A2]	EAB	EAB: Received Invalid Data	Invalid data received from the EAB.	Valid data received from EAB	PTC does not change state	No		EAB Cable, IOC Card, Loco ID Control System Configuration	Yes	No
0x05A8 [L2I474.05A8]	ILC	ILC: Connection Failure	Unable to open or initialize the hardware connection to the ILC (via the IOC).	NA - Unrecoverable Fault	PTC Slice State goes to FAILED	No		EAB Cable, IOC Card, Loco ID Control System Configuration	Yes	No
0x05A9 [L2I474.05A9]	ILC	ILC: Comm Loss	No data being received from the ILC.	Valid data received from ILC	PTC does not change state	No		EAB Cable, IOC Card, Loco ID Control System Configuration	Yes	No
0x05AA [L2I474.05AA]	ILC	ILC: Received Invalid Data	Invalid data being received from the ILC.	Valid data received from ILC	PTC does not change state	No		EAB Cable, IOC Card, Loco ID Control System Configuration	Yes	No
0x05B8 [L2I474.05B8]	LIG	LIG: Connection Failure	Unable to open or initialize the hardware connection	NA – Unrecoverable Fault	PTC Slice State goes to FAILED	No		IOC Cable, TMC, or LIG	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			to the LIG (via the IOC)							
0x05B9 [L2I474.05B9]	LIG	LIG: Comm Loss	Loss of comm between TMC and LIG	Communication restored between TMC and LIG	PTC does not change state	No		IOC Cable, TMC, or LIG	Yes	No
0x05BA [L2I474.05BA]	LIG	LIG: Received Invalid Data	comm problem between TMC and LIG	Communication restored between TMC and LIG	PTC does not change state	No		IOC Cable, TMC, or LIG	Yes	No
0x05D0 [L2I474.05D0]	RSM	RSM: VLAN Config Error	VLAN enabled and VLAN configuration file error exists	VLAN is disabled or VLAN configuration file is correct	PTC does not change state. VLAN will not work	No		CPU Card or Configuration File	Yes	No
0x05E0 [L2I474.05E0]	GPS1	GPS1: S-9103 Time Stamp	GPS1 receives S-9103 messages without required time stamp	GPS1 receives S-9103 messages with required time stamp	Restricts the onboard to operate in Standard Navigation mode instead of Precision Navigation mode	No		GPS	Yes	No
0x05E1 [L2I474.05E1]	GPS2	GPS2: S-9103 Time Stamp	GPS2 receives S-9103 messages without required time stamp	GPS2 receives S-9103 messages with required time stamp	Restricts the onboard to operate in Standard Navigation mode instead of Precision Navigation mode	No		GPS	Yes	No
0x0601	ILC	ILC: Speed	Incoming	Incoming ILC	PTC does	No		ILC, DIO Card, GPS, or	Yes	No

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[L2I474.0601]		Invalid	ILC speed validity bit set to invalid	speed validity bit set to valid	not change state, possible downstream effect of loss of position resulting in change to DISENGAGED			Wheel Tach Sensor		
0x0602 [L2I474.0602]	ILC	ILC: Acceleration Invalid	Incoming ILC acceleration validity bit set to invalid	Incoming ILC acceleration validity bit is set to valid	Degraded Energy Management Functionality	No		ILC	Yes	No
0x0603 [L2I474.0603]	Wheel Tach	Wheel Tach: Speed Invalid	Tachometer input data is invalid/absent	Tachometer input data is present and valid	PTC does not change state, possible downstream effect of loss of position resulting in change to DISENGAGED	No		DIO Card, Wheel Tach Cable, Speed Sensor	Yes	No
0x0604 [L2I474.0604]	CPUx	CPUx: PTC Speed Invalid	When ILC speed is unavailable and any of the following faults are raised: 0x0603, 0x0605, 0x0606, 0x0607.	When ILC speed is unavailable and all of the following faults are clear: 0x0603, 0x0605, 0x0606, 0x0607.	PTC does not change state. Downstream effect of loss of position results in change to DISENGAGED. Degraded Energy Management	No		DIO Card, Wheel Tach Cable, Speed Sensor, ILC, LIG, IOC Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			When ILC speed is available and any of the following faults are raised: 0x0603, 0x0605, 0x0606, 0x0607 as well as any of the following faults are raised: 0x0601, 0x0608, 0x0609, 0x060A, 0x060B	When ILC speed is available and all of the following faults are clear: 0x0603, 0x0605, 0x0606, 0x0607 OR all of the following faults are clear: 0x0601, 0x0608, 0x0609, 0x060A, 0x060B	nt Functionality					
0x0605 [L2I474.0605]	Wheel Tach	Wheel Tach: Sustained Wheel Tach Slip / Slide Detected	Wheel slip and/or slide event lasts for at least 10 seconds and PTC GPS speed is not being used as PTC Speed	Wheel slip and/or slide event lasting for at least 10 seconds is detected as no longer slipping/sliding	PTC does not change state, possible downstream effect of loss of position resulting in change to DISENGAGED	No		DIO Card, GPS Cable, Wheel Tach Cable, Speed Sensor	Yes	No
0x0606 [L2I474.0606]	Wheel Tach	Wheel Tach: Wheel Tach Stuck at Zero	PTC GPS speed greater than 5 mph for at least 10 consecutive seconds while tach	Tach speed is greater than zero	PTC does not change state, possible downstream effect of loss of position	No		DIO Card, Wheel Tach Cable, Speed Sensor	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			speed is zero and PTC GPS speed is not being used as PTC Speed		resulting in change to DISENGAGED					
0x0607 [L2I474.0607]	CPUx	CPUx: Sustained Tach / GPS Speed Source Divergence	More than 20 seconds elapse during which the filtered GPS speed and the filtered Tach speed differ by more than 5 mph	Filtered GPS speed and filtered Tach speed differ by less than 5 mph	PTC does not change state, possible downstream effect of loss of position resulting in change to DISENGAGED	No		DIO Card, GPS Cable, Wheel Tach Cable, Speed Sensor or CPU	Yes	No
0x0608 [L2I474.0608]	ILC	ILC: Sustained ILC Wheel Slip / Slide Detected	ILC Wheel slip and/or slide event lasts for at least 10 seconds.	ILC Wheel slip and/or slide event lasting for at least 10 seconds is detected as no longer slipping/sliding	PTC does not change state, possible downstream effect of loss of position resulting in change to DISENGAGED	No	N/A	IOC Card, ILC	Yes	No
0x0609 [L2I474.0609]	ILC	ILC: ILC Speed Stuck at Zero	PTC GPS speed greater than 5 mph for at least 10 consecutive seconds while ILC speed is zero	ILC speed is greater than zero	PTC does not change state, possible downstream effect of loss of position resulting in change to DISENGAGED	No	N/A	IOC Card, ILC, GPS	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
					ED					
0x060A [L2I474.060A]	CPUx	CPUx: Sustained ILC / GPS Speed Source Divergence	More than 20 seconds elapse during which the filtered GPS speed and the filtered ILC speed differ by more than 5 mph	Filtered GPS speed and filtered ILC speed differ by less than 5 mph	PTC does not change state, possible downstream effect of loss of position resulting in change to DISENGAG ED	No	N/A	IOC Card, ILC, GPS or CPU	Yes	No
0x060B [L2I474.060B]	ILC	ILC: Scaled Speed Invalid	Any of the following faults are raised: 0x0601, 0x0608, 0x0609, 0x060A, 0x0683	All of the following faults are clear: 0x0601, 0x0608, 0x0609, 0x060A, 0x0683	PTC does not change state, possible downstream effect of loss of position resulting in change to DISENGAG ED.	No	N/A	IOC Card, ILC, GPS or CPU	Yes	No
0x060C [L2I474.060C]	Wheel Tach	Wheel Tach: GPS Suspended Sustained Wheel Tach Slip / Slide Detected	Wheel slip and/or slide event lasts for at least 10 seconds and PTC GPS speed is being used as PTC Speed.	Wheel slip and/or slide event lasting for at least 10 seconds is detected as no longer slipping/slidin g or Fault 0x0605 (Wheel Tach: Sustained Wheel Tach Slip / Slide Detected) is raised	PTC does not change state	No		DIO Card, GPS Cable, Wheel Tach Cable, Speed Sensor	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x060D [L2I474.060D]	Wheel Tach	Wheel Tach: GPS Suspended Wheel Tach Stuck at Zero	PTC GPS speed greater than 5 mph for at least 10 seconds while wheel tach speed is zero and PTC GPS speed is being used as PTC Speed	Wheel tach is greater than zero or Fault 0x0606 (Wheel Tach: Wheel Tach Stuck at Zero) is raised	PTC does not change state	No		DIO Card, Wheel Tach Cable, Speed Sensor	Yes	No
0x0614 [L2I474.0614]	TMC	TMC:UTC Time Not Received	RMC packet not received from either GPS receiver within 720 minutes	Valid RMC packet received from either GPS receiver	PTC Slice State goes to FAILED	No		GPS, IOC Cable, or IOC Card	Yes	No
0x0617 [L2I474.0617]	GPS1	GPS1: Data Invalid	GPS 1 data is invalid due to absence or content	Valid data is received from GPS 1	PTC does not change state	No		GPS, IOC Cable, or IOC Card	No	No
0x0618 [L2I474.0618]	GPS2	GPS2: Data Invalid	GPS 2 data is invalid due to absence or content	Valid data is received from GPS 2	PTC does not change state	No		GPS, IOC Cable, or IOC Card	No	No
0x0619 [L2I474.0619]	TMC	TMC: Excessive Antenna Offset Error	TMC detects Excessive Antenna Offset Error	TMC no longer detects Excessive Antenna Offset Error	PTC does not change state	No		Navigation Modules	Yes	No
0x0620 [L2I474.0620]	CPUx	CPUx: Front Brake Pipe Pressure	Loss of comm with braking system for	Communication restored with braking system, and	PTC Slice State goes to FAILED	No		DIO Card, IOC Card, Pressure Sensor, Cable, or Loco ID	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
		Invalid	more than 3 Secs, or pressures between the two sources differ by more than 5 psi during steady state condition	the two sources are within 5 psi for 5 seconds				system data settings		
0x0621 [L2I474.0621]	CPUx	CPUx: Loco Brake Cylinder Pressure Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No	For Control System Types: ACP, LIG, LSI, QES, TMC-DIO.	IOC Card, Cable, or Loco ID system data settings	Yes	No
0x0622 [L2I474.0622]	CPUx	CPUx: Brake Cylinder Commanded Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x0623 [L2I474.0623]	CPUx	CPUx: Equalizing Reservoir Pressure Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	PTC does not change state	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x0625 [L2I474.0625]	CPUx	CPUx: Feed Valve Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x0626 [L2I474.0626]	CPUx	CPUx: Bail Indication Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0628 [L2I474.0628]	CPUx	CPUx: Air Brake Cut In Status Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	PTC Slice State goes to FAILED	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x0629 [L2I474.0629]	CPUx	CPUx: Air Brake Setup Indication Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x062A [L2I474.062A]	CPUx	CPUx: Elec Air brake Fault Status Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	PTC does not change state	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x062B [L2I474.062B]	CPUx	CPUx: Auto Brake Handle Position Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x062C [L2I474.062C]	CPUx	CPUx: Independent Handle Position Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x062D [L2I474.062D]	CPUx	CPUx: Penalty Applied Indication Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC does not change state	No		EBI Card	Yes	No
0x062E [L2I474.062E]	CPUx	CPUx: Penalty Cutout Indication	Invalid message from EBI or non comm	Communication restored with EBI	PTC Slice State goes to FAILED	No		EBI Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
		Invalid	with EBI for 3 seconds							
0x062F [L2I474.062F]	CPUx	CPUx: Penalty Drive Status Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0630 [L2I474.0630]	CPUx	CPUx: Penalty Brake Source Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0631 [L2I474.0631]	CPUx	CPUx: Emer Brake Applied Status Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC does not change state	No		EBI Card	Yes	No
0x0632 [L2I474.0632]	CPUx	CPUx: Emer Brake Cutout Indication Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0633 [L2I474.0633]	CPUx	CPUx: Emer Brake Drive Status Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0634 [L2I474.0634]	CPUx	CPUx: Emer Brake Source Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC does not change state	No		EBI Card	Yes	No
0x0635	CPUx	CPUx: Horn	Invalid	Communication	PTC Slice	No		EBI Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0635]		Cutout Indication Invalid	message from EBI or non comm with EBI for 3 seconds	on restored with EBI	State goes to FAILED					
0x0636 [L2I474.0636]	CPUx	CPUx: Horn Drive Status Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0637 [L2I474.0637]	CPUx	CPUx: Simplex Status Indication Invalid	Invalid message from EBI or non comm with EBI for 3 seconds	Communication restored with EBI	PTC Slice State goes to FAILED	No		EBI Card	Yes	No
0x0638 [L2I474.0638]	CPUx	CPUx: Brake Pipe Flow Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x0639 [L2I474.0639]	CPUx	CPUx: Main Reservoir Pressure Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x063A [L2I474.063A]	CPUx	CPUx: ER Target Pressure Invalid	Loss of comm with braking system for more than 3 Secs	Communication restored with braking system	Degraded Energy Management Functionality	No		IOC Card, Cable, or Loco ID system data settings	Yes	No
0x063D [L2I474.063D]	Discrete Interface	Discrete Interface: Atmosphere Vent Pressure	ATM/Vent Sensor pressure is greater than 3 psi or less	ATM/Vent Sensor pressure is between -3 and 3 psi	PTC does not change state.	No	This fault is only raised on control systems where the vent sensor is required.	DIO Card, ATM/Vent Sensor, DIO Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
		Invalid	than -3 psi	inclusive			For Control System Types: LIG: This fault is raised if CFG50 is Enabled or Disabled.			
0x0640 [L2I474.0640]	EOT	EOT: EOT Brake Pipe Pressure Invalid	When the data is not received from the data source or BPP exceeds 135 psi	When data received from the data source and BPP is less than or equal to 135 psi	PTC does not change state, braking algorithm is more conservative	No		EOT, HOT, DIO Card, or Cables	Yes	No
0x0641 [L2I474.0641]	EOT	EOT: EOT Radio Freq No Comm Invalid	Message from HOT indicates no comm	Message from HOT indicates comm	PTC does not change state	No		EOT, HOT, DIO Card, or Cables	Yes	No
0x0650 [L2I474.0650]	Discrete Interface	Discrete Interface: Reverser Position Invalid	When the Reverser Position is invalid and PTC is not maintaining a Forward or Reverse Reverser Position due to being moving and in an Active, Disengaged or Restricted state.	Reverser Position Valid	PTC Slice State goes to FAILED	No		DIO Card, LIG, or Cable	Yes	No
0x0651 [L2I474.0651]	Discrete Interface	Discrete Interface: Throttle Notch Invalid	Invalid throttle valve combination is detected	Current valid throttle valve combination	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No

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0x0652 [L2I474.0652]	Discrete Interface	Discrete Interface: Engine Run Indication Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for 3 seconds	Valid message from DIO/LIG	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No
0x0653 [L2I474.0653]	Discrete Interface	Discrete Interface: Generator Field Indication Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No
0x0654 [L2I474.0654]	Discrete Interface	Discrete Interface: Traction Motor Current Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds. Invalid if traction type is set to AC	Valid message from DIO/LIG	Degraded Energy Management Functionality	No		DIO Card, or Cable	No	No
0x0655 [L2I474.0655]	Discrete Interface	Discrete Interface: Generator Voltage Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No
0x0656 [L2I474.0656]	Discrete Interface	Discrete Interface: Tractive Effort	Invalid message from DIO/LIG or	Valid message from DIO/LIG	Degraded Energy Management	No		DIO Card, or Cable	Yes	No

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	e	Invalid	non comm with DIO/LIG for more than 3 seconds. Invalid if traction type is set to DC		Functionalit y					
0x0657 [L2I474.0657]	Discret e Interfac e	Discrete Interface: Dynamic Brake Setup Status Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Manageme nt Functionalit y	No		DIO Card, or Cable	Yes	No
0x0658 [L2I474.0658]	Discret e Interfac e	Discrete Interface: DB Start Status Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Manageme nt Functionalit y	No		DIO Card, or Cable	Yes	No
0x0659 [L2I474.0659]	Discret e Interfac e	Discrete Interface: DB On Status Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Manageme nt Functionalit y	No		DIO Card, or Cable	Yes	No
0x065A [L2I474.065A]	Discret e Interfac e	Discrete Interface: DB Excitation Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for	Valid message from DIO/LIG and voltage is between 0V and 74V	Degraded Energy Manageme nt Functionalit y	No		DIO Card, or Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			more than 3 seconds. Invalid if voltage is less than 0V or greater than 74V							
0x065B [L2I474.065B]	Discrete Interface	Discrete Interface: Power Cutoff Solenoid Open Invalid	Invalid message from DIO or non comm with DIO for more than 3 seconds	Valid message from DIO	PTC Slice State goes to FAILED	No		DIO Card, or Cable	Yes	No
0x065C [L2I474.065C]	Discrete Interface	Discrete Interface: EC Switch Position Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds. Invalid if "Isolate" and "Run" show high at same time	Valid message from DIO/LIG and either "Isolate" or "Run" show high at any point in time	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No
0x065D [L2I474.065D]	Discrete Interface	Discrete Interface: Wheel Slip Indication Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No
0x065E [L2I474.065E]	Discrete Interface	Discrete Interface: Battery Knife Switch Status	Invalid message from DIO/LIG or non comm	Valid message from DIO/LIG	PTC does not change state	No		DIO Card, or Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
		Invalid	with DIO/LIG for more than 3 seconds							
0x065F [L2I474.065F]	Discrete Interface	Discrete Interface: Sanding Indication Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No
0x0660 [L2I474.0660]	Discrete Interface	Discrete Interface: Emergency Sanding Indication Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No
0x0661 [L2I474.0661]	EBI	EBI: Horn Indication Invalid	Invalid message from EBI or non comm with EBI for more than 3 seconds	Valid message from EBI	Impact to PTC horn activation function, displays "HORN" flag, does not change state	No		EBI Card, or Cable	Yes	No
0x0662 [L2I474.0662]	Discrete Interface	Discrete Interface: Bell Indication Invalid	Invalid message from DIO/LIG or non comm with DIO/LIG for more than 3 seconds	Valid message from DIO/LIG	Degraded Energy Management Functionality	No		DIO Card, or Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0663 [L2I474.0663]	Locomotive	Locomotive: Battery knife Switch Open	Battery knife switch is in open state	Battery knife switch is in closed state	PTC does not change state. Downstream effects may cause state to become DISENGAGED.	No		Battery knife switch, Power cable	Yes	No
0x0669 [L2I474.0669]	CSM	CSM: Cab Code 270/270 Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds. When this signal input is showing high with any other signal indication or enabled in the LocoID	Valid message from IOC and signal input is not high with any other signal indication	PTC State does not change state, system assumes indication is Restricting	No		IOC Card, or IOC Cable	Yes	No
0x0670 [L2I474.0670]	CSM	CSM: Cab Code 180 Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds. When this signal input is showing high with any other signal indication or enabled in the LocoID	Valid message from IOC and signal input is not high with any other signal indication	PTC State does not change state, system assumes indication is Restricting	No		IOC Card, or IOC Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0671 [L2I474.0671]	CSM	CSM: Cab Code 120 Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds. When this signal input is showing high with any other signal indication or enabled in the LocoID	Valid message from IOC and signal input is not high with any other signal indication	PTC State does not change state, system assumes indication is Restricting	No		IOC Card, or IOC Cable	Yes	No
0x0672 [L2I474.0672]	CSM	CSM: Cab Code 75 Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds. When this signal input is showing high with any other signal indication or enabled in the LocoID	Valid message from IOC and signal input is not high with any other signal indication	PTC State does not change state, system assumes indication is Restricting	No		IOC Card, or IOC Cable	Yes	No
0x0673 [L2I474.0673]	CSM	CSM: Cab Code 0 Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds. When this signal input is showing high with	Valid message from IOC and signal input is not high with any other signal indication	PTC State does not change state, system assumes indication is Restricting	No		IOC Card, or IOC Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			any other signal indication or enabled in the LocoID							
0x0674 [L2I474.0674]	CSM	CSM: Cut Out Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds	Valid message from IOC	PTC State does not change state.	No		IOC Card, or IOC Cable	Yes	No
0x0675 [L2I474.0675]	CSM	CSM: Cab Code 270 Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds. When this signal input is showing high with any other signal indication or enabled in the LocoID	Valid message from IOC and signal input is not high with any other signal indication	PTC State does not change state, system assumes indication is Restricting	No		IOC Card, or IOC Cable	Yes	No
0x0676 [L2I474.0676]	CSM	CSM: Discrete and Raw Signals Mismatch	Raw and Discrete messages indicate signal indications mismatch for more than 3 seconds tolerance	Raw and Discrete messages indicate matching signal indications	PTC State does not Change state, system assumes indication is Restricting	No		IOC Card	No	No
0x0677	CSM	CSM: Cab Code 75/75 Indication	Invalid message from IOC or	Valid message from IOC and	PTC State does not change	No		IOC Card, or IOC Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0677]		Invalid	non comm with IOC for more than 3 seconds. When this signal input is showing high with any other signal indication or enabled in the LocOLD	signal input is not high with any other signal indication	state, system assumes indication is Restricting					
0x0678 [L2I474.0678]	CSM	CSM: Cab Code 120/120 Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds. When this signal input is showing high with any other signal indication or enabled in the LocOLD	Valid message from IOC and signal input is not high with any other signal indication	PTC State does not change state, system assumes indication is Restricting	No		IOC Card, or IOC Cable	Yes	No
0x0679 [L2I474.0679]	CSM	CSM: Cab Code 180/180 Indication Invalid	Invalid message from IOC or non comm with IOC for more than 3 seconds. When this signal input is showing high with any other signal indication or	Valid message from IOC and signal input is not high with any other signal indication	PTC State does not change state, system assumes indication is Restricting	No		IOC Card, or IOC Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			enabled in the LocoID							
0x067A [L2I474.067A]	CSM	CSM: Serial Cab Signal Comm Loss	When Serial Cab Signal System Interface enabled and no valid message from Serial Cab Signal System for more than 3 seconds	Valid message from Serial Cab Signal System	PTC State does not change state, system assumes Higher Speed Operation Disabled	No		IOC Card, or IOC Cable	Yes	No
0x0680 [L2I474.0680]	EBI	EBI: Card Fault Horn Circuit Failure	Invalid data received from EBI	Valid data received from EBI	PTC Slice State goes to FAILED	No		EBI Card, or Cable	Yes	No
0x0681 [L2I474.0681]	TMC	TMC: Wheel Calibration Fault	Previously calculated valid wheel diameter cannot be loaded at powerup or currently loaded / calculated wheel diameter is greater than 105% or less than 85% of the LocoID module wheel diameter	Currently loaded / calculated wheel diameter is within 105% to 85% of the LocoID module wheel diameter	PTC defaults to LocoID module wheel diameter if valid wheel diameter is not available from any CPU	No		LocoID setting for PPR and Wheel Diameter or Wheel calibration needs to be run	Yes	No
0x0682 [L2I474.0682]	EBI	EBI: Horn Circuit Input Fault	Invalid data input to EBI from horn interface	Valid data input to EBI from horn interface	PTC displays "HORN" flag, does not change	No		Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			wiring	wiring	state					
0x0683 [L2I474.0683]	TMC	TMC: ILC Scaling Calibration Fault	Previously calculated ILC Scaling Factor cannot be loaded at powerup or currently loaded / calculated ILC Scaling Factor is outside the range of 0.9375 to 1.0666	Currently loaded / calculated ILC Scaling Factor is within the range of 0.9375 to 1.0666	ILC Speed unavailable for use as input to PTC Speed. PTC does not change state, possible downstream effect of loss of position resulting in change to DISENGAGED	No	N/A	IOC Card, ILC, GPS or CPU	Yes	No
0x0690 [L2I474.0690]	CPUx	CPUx: Brake Cylinder Pressure Invalid	Loss of comm between DIO and CPU for more than 3 Secs or Brake Cylinder Pressure is out of range	Communication restored between DIO and CPU and Brake Cylinder Pressure is in range	PTC Slice State goes to FAILED	Yes	Applies to Control System Types NS 300 EMU, NS HL2 EMU, NS MC EMU, NS MCEW EMU, SR FLRT3 DMU, SR KSS EMU	DIO Card, BCP sensor, DIO Cable	Yes	No
0x0691 [L2I474.0691]	Discrete Interface	Discrete Interface: Cab Control Invalid	Loss of comm with DIO for more than 3 Secs or Cabin Control Signal combination invalid	Communication restored with DIO and Cabin Control Signal combination is valid	PTC Slice State goes to FAILED	Yes	Applies to Control System Types SR FLRT3 DMU, NS MC EMU, NS MCEW EMU	DIO, CPU Card, DIO Cable	Yes	No
0x0692	Discrete Interface	Discrete Interface: Motor Braking	Loss of comm between	Communication restored between DIO	PTC Slice State goes	Yes	Applies to Control System Types NS 300	DIO, CPU Card, DIO Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0692]	e	PWM Invalid	DIO and CPU for more than 3 Secs or the Motor Braking PWM input combination is invalid	and CPU and the Motor Braking PWM input combination is valid	to FAILED		EMU and NS HL2 EMU. NS 300 EMU uses the combination of PWM1 and PWM2. NS HL2 EMU uses the combination of Air Brake Cut-In, Penalty Brake Applied, PWM1 and PWM2.			
0x0693 [L2I474.0693]	Discrete Interface	Discrete Interface: Brake Effort Invalid	Loss of comm between DIO and CPU for more than 3 Secs or Brake Effort is out of Range	Communication restored between DIO and CPU and Brake Effort is in range	PTC does not change state	No	Applies to Control System Type SR FLRT3 DMU	DIO Card, DIO Cable, or Loco ID system data settings	Yes	No
0x0694 [L2I474.0694]	Discrete Interface	Discrete Interface: Brake Demand Invalid	Loss of comm between DIO and CPU for more than 3 Secs or Brake Demand is out of Range	Communication restored between DIO and CPU and Brake Demand is in Range	PTC does not change state	No	Applies to Control System Type NS HL2 EMU and SR FLRT3 DMU. For NS HL2 EMU uses the combination of PTC Controlling, PWM1 and PWM2, Brake Demand. For SR FLRT3 DMU uses the combination of BCP, Brake Demand, Traction Demand and Speed.	DIO Card, DIO Cable, or Loco ID system data settings	Yes	No
0x0695 [L2I474.0695]	Discrete Interface	Discrete Interface: Traction Demand Invalid	Loss of comm between DIO and CPU for more than 3 Secs or Traction	Communication restored between DIO and CPU and Traction Demand is in Range	PTC does not change state	No	Applies to Control System Type SR FLRT3 DMU	DIO Card, DIO Cable, or Loco ID system data settings	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			Demand is out of Range							
0x0696 [L2I474.0696]	Discrete Interface	Discrete Interface: Tractive Effort Invalid	Loss of comm between DIO and CPU for more than 3 Secs or Tractive Effort is out of range	Communication restored between DIO and CPU and Tractive Effort is in range	PTC does not change state	No	Applies to Control System Type SR FLRT3 DMU	DIO Card, DIO Cable, or Loco ID system data settings	Yes	No
0x0697 [L2I474.0697]	Discrete Interface	Discrete Interface:	Loss of comm between DIO and CPU for more than 3 Secs or Brake Feedback PWM Range Invalid	Communication restored between DIO and CPU and Brake Feedback PWM is in range	PTC does not change state	No	Applies to Control System Types NS 300 EMU, NS HL2 EMU	DIO Card, DIO Cable, or Loco ID system data settings	Yes	No
0x0698 [L2I474.0698]	Discrete Interface	Discrete Interface: EP pressure switch invalid	Loss of comm between DIO and CPU for more than 3 Secs or EP pressure switch is invalid	Communication restored between DIO and CPU and EP pressure switch is valid	PTC does not change state	No	Applies to Control System Types NS 300 EMU, NS HL2 EMU, SR FLRT3 DMU	DIO, CPU Card, DIO Cable, EP Pressure Switch	Yes	No
0x0699 [L2I474.0699]	DIO	DIO: Demand1/Demand2 Mismatch	Difference between Demand1 setting and Demand2 setting > 5% during	Demand setting difference between Demand1 and Demand2 is	PTC State Change goes to FAILED	No	Applies to Control System Type SR KSS EMU	DIO Card, Demand Circuits AI4 and or AI6, DIO Cable	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			steady state condition	< 5%.						
0x069A [L2I474.069A]	DIO	DIO: BCP1 Cut In Open	Cabinet BCP1 valve is open	Cabinet BCP1 valve is closed	PTC State does not change state	No	Applies to Control System Type SR KSS EMU	DIO Card, BCP sensor, DIO Cable	Yes	No
0x069B [L2I474.069B]	Discrete Interface	Discrete Interface: EPPS invalid	Emergency Pipe Pressure Switch (EPPS) is invalid	EPPS is valid	PTC Slice State goes to FAILED	No	Control Type: SR KSS EMU Application: EB Loop (vehicle control loop for emergency brake application) and MBP Status (pneumatic sense of the emergency magnet valve state (Closed/Open)) are tied logically to the EPPS. If EPPS is invalid, then ability to command an emergency brake application is at risk.	DIO, CPU Card, DIO Cable, EP Pressure Switch	Yes	No
0x0700 [L2I474.0700]	TMC	TMC: Multiple CPU Faults	More than one CPU declared faulty	At least two CPUs are functional	PTC Slice State goes to FAILED	No		Power cycle the TMC, Update Onboard software, or TMC	Yes	No
0x0702 [L2I474.0702]	TMC	TMC: Multiple CPU Faults	Both Secondary CPUs have been voted out and only the Primary CPU is operating (vitality is lost) and not all conditions to	At least one Secondary CPU is operating and has been voted back in – vitality is restored	PTC Slice State goes to FAILED	Yes		CPU	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			trigger the 0x0819 fault are met							
0x0703 [L2I474.0703]	CPUx	CPUx: CPU-1 Voted-out	Vote-out message is sent to EBI indicating CPU1 data does not match peers	This fault is cleared when the vote-out slice's data is matched with other peers. This can also be cleared by re-cycling the power of TMC.	PTC does not change state	No		CPU or EBI Card	Yes	Yes
0x0704 [L2I474.0704]	CPUx	CPUx: CPU-2 Voted-out	Vote-out message is sent to EBI indicating CPU2 data does not match peers	This fault is cleared when the vote-out slice's data is matched with other peers. This can also be cleared by re-cycling the power of TMC.	PTC does not change state	No		CPU or EBI Card	Yes	Yes
0x0705 [L2I474.0705]	CPUx	CPUx: CPU-3 Voted-out	Vote-out message is sent to EBI indicating CPU3 data does not match peers	This fault is cleared when the vote-out slice's data is matched with other peers. This can also be cleared by re-cycling the power of TMC.	PTC does not change state	No		CPU or EBI Card	Yes	Yes
0x0706 [L2I474.0706]	CPUx	FC Unexpected Peer	If a CPU is communicating with comparator that is not	NA – Fault will remain active	PTC does not change state	No		CPU	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			expected							
0x0707 [L2I474.0707]	TMC	TMC: Self-Test Timeout	The on-board segment is performing file compression and the System State remains in Self-Test for longer than 90 minutes. Or The on-board segment is not performing file compression and the System State remains in Self-Test for longer than 20 minutes.	"Power Cycle" the TMC	PTC Slice State goes to Failed	Yes		None	Yes	No
0x0708 [L2I474.0708]	CPUx	CPUx: Object in Resync	Data Manager Initialization Sync Request Received (object startup for those objects not	Target Generator Object Ready to Generate Targets (exact conditions vary by	None	No	This fault prevents a slice from exiting resync until all target generator objects are ready to generate their targets	N/A	No	Yes

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			registered to receive the Data Manager callback)	object)						
0x0709 [L2I474.0709]	CPUx	CPUx: In Self Test	CPU in Self Test state	CPU in any state other than Self Test	PTC does not change state	No	This fault does not represent a failure of the on-board segment. The on-board segment will not allow a transition to the Initializing state unless all three application CPUs are available.	None	No	Yes
0x070A [L2I474.070A]	CPUx	CPUx: Loss of Availability During Initialization	A change in which CPU is Primary while the system is in the INITIALIZING state	Cleared upon transition to FAILED	PTC Slice State goes to FAILED	No		Onboard Software on CPU or CPU Card	Yes	No
0x0710 [L2I474.0710]	CPUx	CPUx: Navigation Fault	Onboard has one or more solutions for current location but none has a CTE within the max limits	Onboard has at least one solution for current location with its CTE within limits	PTC does not change state. Downstream effects may cause state to become DISENGAGED.	No		GPS, or Track Database	Yes	No
0x0711 [L2I474.0711]	CPUx	CPUx: Navigation Fault	Onboard has an unknown track location in the ACTIVE state	Onboard has a known track location or System State Becomes DISENGAGED	PTC does not change state. Downstream effects may cause state to become DISENGAGED	No		GPS, or Track Database	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
					ED.					
0x0712 [L2I474.0712]	CPUx	CPUx: Max Position Uncertainty Reached	When along track position uncertainty reaches its max value TBC201	Along track position uncertainty is below the max value TBC201	PTC Slice State goes DISENGAGED	No		GPS	Yes	No
0x0720 [L2I474.0720]	CPUx	CPUx: Software Fault	Operating System has run out of timer resources so a new timer cannot be created.	Other timers have been destroyed and resources are available to create a new timer.	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0721 [L2I474.0721]	CPUx	CPUx: Software Fault	The Data Manager Object has run out of memory and cannot add anything more to the triplex store, or a user of the triplex store encounters a failure to access critical information from the triplex store	Memory has become available and the data item was created or stored, and accessed successfully	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0722 [L2I474.0722]	CPUx	CPUx: Software Fault	The Display Manager Object has run out of memory and cannot store the callback	Memory has become available and the callback function was added to the	PTC Slice State goes to FAILED	No		CPU Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			function.	list.						
0x0730 [L2I474.0730]	CPUx	CPUx: Invalid Message From BOS	Internal data error	NA - Fault is raised on an individual message basis	PTC does not change state	No		CPU Card	Yes	No
0x0731 [L2I474.0731]	CPUx	CPUx: Software Fault	The Data Manager Object was not able to retrieve the data item from the triplex store.	The item was retrieved properly from the triplex store.	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0732 [L2I474.0732]	CPUx	CPUx: Software Fault	On-Board is unable to create a target when it believes a target is required.	Expected targets are successfully generated (Transition to FAILED removes the need for most targets)	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0733 [L2I474.0733]	CPUx	CPUx: Software Fault	Internal Buffer Overflow	The Internal Buffer has space available. (May require a power cycle)	PTC Slice State goes to FAILED	Yes		CPU Card	Yes	No
0x0734 [L2I474.0734]	CPUx	CPUx: Software Fault	The Data Manager Object has run out of memory and cannot add anything more to the triplex store.	Memory has become available and the data item was created or stored successfully.	PTC does not change state	No		CPU Card	Yes	No
0x0735	CPUx	CPUx: PTC	There is a	Sixty (60)	PTC Slice	No	This can occur, for	None	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0735]		suspension splits signal block	signal block split by one or more maximum authorized speed PTC suspension targets into multiple unsuspended areas.	seconds have elapsed since the fault was raised.	State goes FAILED		example, when the start or end of a maximum authorized PTC suspension target is not collocated with a facing signal, an opposing signal, a clearing or signal in lieu of electric lock clearance point, or a transition from non-PTC track /PTC Entry track/ PTC track not signaled in the train's direction of movement to PTC track signaled in the train's direction of movement			
0x0740 [L2I474.0740]	CPUx	CPUx: Software Fault	Software object problem	Re-Initialize the TMC. (Go through INIT)	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0741 [L2I474.0741]	CPUx	CPUx: Software Fault	Braking calculation fails	1) When a valid reactive braking curve is calculated and the locomotive is stopped, or 2) When the On-board state is non-controlling	PTC does not immediately change state, but will transition to DISENGAGED after acknowledgment of degraded system state prompt.	No		CPU Card	Yes	No
0x0743 [L2I474.0743]	CPUx	CPUx: Software Fault	Three seconds elapsed since the enforcement object (which	1) When a valid reactive braking curve is calculated and the locomotive is	PTC does not immediately change state, but will transition to	No		CPU Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			evaluates targets against the braking curve) last received a braking curve and the locomotive track location and direction of movement are known.	stopped, or 2) When the On-board state is non-controlling	DISENGAGED after acknowledgment of degraded system state prompt.					
0x0747 [L2I474.0747]	CPUx	CPUx: Software Fault	When there is a failure retrieving consist data and the On-board state is active and the train is moving.	Valid consist data is retrieved	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0748 [L2I474.0748]	CPUx	CPUx: Software Fault	If the Target Library is not able to create a target	Clears when the target is deleted by the Target Library that called for its creation	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0749 [L2I474.0749]	Discrete Interfaces	Discrete Interface: Throttle Data Invalid	Throttle Data is invalid	Throttle Data is valid	System assumes a valid maximum throttle setting to ensure a longer (more conservative) warning	No	Applies to Control System Types ACP, LIG, LSI, QES, TMC-DIO	DIO Card or LIG	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
					time					
0x074A [L2I474.074A]	CPUx	Potential Excessive Unmapped Grade	A portion of the train is on an exclusion zone, or on an unmapped track, and the mapped track under the other portion of the train has a grade greater than $ +/- 1.25\% $	Any part of the train is not on an exclusion zone, or all of the train is on mapped track, or a portion of the train is on unmapped track and the mapped track under the other portion of the train has a grade that is $ +/- 1.25\% $ or less	PTC does not change state	No		N/A	Yes	No
0x074B [L2I474.074B]	Discrete Interface	Discrete Interface: Reverser Position Data Invalid	Invalid message from DIO or LIG or non comm with DIO or LIG or Reverser Input Signals invalid	Valid message from DIO or LIG and Reverser Input Signals valid	PTC does not change state	No		DIO Card, LIG, or Cable	Yes	No
0x0760 [L2I474.0760]	Track Data	Track Data: Invalid Data	Track manager is unable to open a track file or finds one of the track header or feature size out of limits	When the system state transits to INITIALIZATION	PTC does not change state, but will not use affected track database	No		Track Database	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0770 [L2I474.0770]	CPUx	CPUx: Invalid Data	Invalid Consist Data received from Triplex Data Store	Initialize On-board, On-board state is Failed, or receive valid 1030 message from BOS	PTC Slice State goes to FAILED	No		CPU Card	Yes	No
0x0780 [L2I474.0780]	CDU1	CDU1: Software Fault	Unable to open or initialize the hardware connection to the local diagnostics server on CDU1.	Hardware connection to the local diagnostics server on CDU1 is opened	PTC State does not Change	No		CDU1	Yes	No
0x0781 [L2I474.0781]	CDU1	CDU1: Software Fault	Invalid data received from the local diagnostics server on CDU1.	Valid data received from the local diagnostics server on CDU1	PTC State does not Change	No		CDU1	Yes	No
0x0782 [L2I474.0782]	CDU1	CDU1: Software Fault	No data received from the local diagnostics server on CDU1.	Data received from the local diagnostics server on CDU1	PTC State does not Change	No		CDU1	Yes	No
0x0783 [L2I474.0783]	CDU2	CDU2: Software Fault	Unable to open or initialize the hardware connection to the local diagnostics server on CDU2.	Hardware connection to the local diagnostics server on CDU2 is opened	PTC does not change state	No		CDU2	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0784 [L2I474.0784]	CDU2	CDU2: Software Fault	Invalid data received from the local diagnostics server on CDU2.	Valid data received from the local diagnostics server on CDU2	PTC does not change state	No		CDU2	Yes	No
0x0785 [L2I474.0785]	CDU2	CDU2: Software Fault	No data received from the local diagnostics server on CDU2.	Data received from the local diagnostics server on CDU2	PTC does not change state	No		CDU2	Yes	No
0x0790 [L2I474.0790]	Config File	Config File: missing railroads.cfg	No railroads.cfg file is available	A railroads.cfg file becomes available	PTC Slice State goes to FAILED	No		Ensure that there is a railroads.cfg file in the common configuration fileset(s) in the owning railroad MDM repository	Yes	No
0x0792 [L2I474.0792]	CDU1	CDU1: Class D Connection Failed	Software object problem	Class D connection restored	PTC State does not Change	No		Update railroad.cfg under usr/share/config/common/ folder, or CDU	Yes	No
0x0793 [L2I474.0793]	CDUx	CDUx: Disk Space on CDU is full	Disk space on CDU exceeds space limit	Disk space on CDU is within space limits	PTC State does not Change	No		CDU	Yes	No
0x0800 [L2I474.0800]	TMC	TMC: Software Fault	Indicates that a software object included in the startup files has died and been restarted and not all conditions to trigger the	This fault is cleared after a set amount of time long enough to indicate that the restart has occurred	PTC State Goes to Failed	Yes		CPU Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			0x081A fault are met							
0x0801 [L2I474.0801]	TMC	TMC: Internal Comm Fault	A set period of time has passed without receiving any responses to periodic self test messages from the software objects included in the startup scripts	A response is received for a periodic self test message from one of the software objects included in the startup scripts	PTC State does not Change	No		CPU Card	Yes	No
0x0802 [L2I474.0802]	TMC	TMC: External Comm Fault	A set period of time has passed without receiving any responses to periodic self test messages between the test manager running on the source component and the test managers running on all of the other components	A response is received for a periodic self test message between the test manager on the source component and one of the test managers running on another component	PTC State does not Change	No		CPU Card, or CDU	Yes	No
0x0803	TMC	TMC: Software	One or more of the software	All software objects included in	PTC State does not	No		CPU Card, or CDU	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0803]		Fault	objects included in the startup scripts are not currently running or is not responding to periodic self test messages	the startup scripts are currently running and responding to periodic self test messages	Change					
0x0805 [L2I474.0805]	TMC	TMC: Software Fault	Files corrupt or missing with application software	Valid application software files	PTC Slice State goes to FAILED	No		CPU Card, or CDU	Yes	No
0x0806 [L2I474.0806]	TMC	TMC: Software Version Mismatch	There is a software version mismatch between any of the following components : CPU1, CPU2, CPU3, CDU1, CDU2	There is no software version mismatch between the following components: CPU1, CPU2, CPU3, CDU1, CDU2	PTC State does not Change	No		Reload Software, CPU Card, CDU	Yes	No
0x0807 [L2I474.0807]	TMC	TMC: Software Version Mismatch	The PTC State is DISENGAGED, ACTIVE or RESTRICTED and there is a software version mismatch between any of the following	The fault has been raised for at least 10 seconds	PTC Slice State goes to FAILED	No		Reload Software, CPU Card, CDU	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			components : CPU1, CPU2, CPU3, CDU1, CDU2							
0x0808 [L2I474.0808]	CPUx	CPUx: OS not compatible	Current operating system is not 'QOS'	NA – Unrecoverable Fault	PTC Slice State goes to FAILED	No		CPU Card, or SW	Yes	Yes
0x0810 [L2I474.0810]	TMC	TMC: Comm Fault with CPU1	The Test Manager object on the source component has lost communication with the Test Manager object on CPU1	The Test Manager object on the source component has restored communication with the Test Manager object on CPU1	PTC State does not Change	No		CPU Card, or CDU	Yes	Yes
0x0811 [L2I474.0811]	TMC	TMC: Comm Fault with CPU2	The Test Manager object on the source component has lost communication with the Test Manager object on CPU2	The Test Manager object on the source component has restored communication with the Test Manager object on CPU2	PTC State does not Change	No		CPU Card, or CDU	Yes	Yes
0x0812 [L2I474.0812]	TMC	TMC: Comm Fault with CPU3	The Test Manager object on the source component has lost communication with the	The Test Manager object on the source component has restored communication with the	PTC State does not Change	No		CPU Card, or CDU	Yes	Yes

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			Test Manager object on CPU3	Test Manager object on CPU3						
0x0815 [L2I474.0815]	TMC	TMC: Comm Fault with CDU1	The Test Manager object on the source component has lost communication with the Test Manager object on CDU1	The Test Manager object on the source component has restored communication with the Test Manager object on CDU1	PTC State does not Change	No		CPU Card, or CDU	Yes	No
0x0816 [L2I474.0816]	TMC	TMC: Comm Fault with CDU2	The Test Manager object on the source component has lost communication with the Test Manager object on CDU2	The Test Manager object on the source component has restored communication with the Test Manager object on CDU2	PTC State does not Change	No		CPU Card, or CDU	Yes	No
0x0817 [L2I474.0817]	TMC	TMC: Out Of Memory	Internal memory usage attempts to exceed memory usage limit	TMC Power Cycle	PTC Slice State goes to FAILED	No		CPU Card, or CDU	Yes	No
0x0818 [L2I474.0818]	TMC	TMC: CPU Usage Too High	Single process is greater than 10% and total CPU usage is greater than	Total CPU usage is less than 95%	PTC Slice State goes to FAILED if On-board segment state is ACTIVE or	No		CPU Card, or SW	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			99% for at least 4 seconds		RESTRICTED					
0x0819 [L2I474.0819]	TMC	TMC: Multiple CPU Faults	Crew is logged off and both Secondary CPUs have been voted out and only the Primary CPU is operating - no automatic reboot of CPU	At least one Secondary CPU is operating and has been voted back in	PTC Slice State goes to FAILED	No		CPU	Yes	No
0x081A [L2I474.081A]	TMC	TMC: Software Fault	Indicates that a software object included in the startup files has died and been restarted while the crew is logged off	This fault is cleared after a set amount of time long enough to indicate that the restart has occurred	PTC State Goes to FAILED	No		CPU Card	Yes	No
0x0851 [L2I474.0851]	TMC	TMC: Failure to Connect to CPU1	Unable to open or initialize the hardware connection to the local diagnostics server on CPU1.	Hardware connection to the local diagnostics server on CPU1 is opened	PTC does not change state	No		CPU Card or SW	Yes	Yes
0x0852	TMC	TMC: Failure to Connect to	Unable to open or initialize the	Hardware connection to the local	PTC does not change	No		CPU Card or SW	Yes	Yes

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
[L2I474.0852]		CPU2	hardware connection to the local diagnostics server on CPU2.	diagnostics server on CPU2 is opened	state					
0x0853 [L2I474.0853]	TMC	TMC: Failure to Connect to CPU3	Unable to open or initialize the hardware connection to the local diagnostics server on CPU3.	Hardware connection to the local diagnostics server on CPU3 is opened	PTC does not change state	No		CPU Card or SW	Yes	Yes
0x0854 [L2I474.0854]	TMC	TMC: Failure to Connect to CPU4	Unable to open or initialize the hardware connection to the local diagnostics server on CPU4.	Hardware connection to the local diagnostics server on CPU4 is opened	PTC does not change state	No		CPU Card or SW	Yes	No
0x0855 [L2I474.0855]	TMC	TMC: Failure to Connect to RSM	Unable to open or initialize the hardware connection to the local diagnostics server on RSM.	Hardware connection to the local diagnostics server on RSM is opened	PTC does not change state	No		RSM Card	Yes	No
0x0857 [L2I474.0857]	TMC	TMC: Failure to Connect to DIO	Unable to open or initialize the hardware connection to the local diagnostics server on	Hardware connection to the local diagnostics server on DIO is opened	PTC does not change state	No		DIO Card	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			DIO.							
0x0858 [L2I474.0858]	TMC	TMC: Failure to Connect to IOC	Unable to open or initialize the hardware connection to the local diagnostics server on IOC.	Hardware connection to the local diagnostics server on IOC is opened	PTC does not change state	No		IOC Card	Yes	No
0x085A [L2I474.085A]	TMC	TMC: Failure to Connect to CPU10	Unable to open or initialize the hardware connection to the local diagnostics server on CPU10.	Hardware connection to the local diagnostics server on CPU10 is opened	PTC does not change state	No		CPU Card or SW	Yes	No
0x85B [L2I474.085B]	CPUx	CPUx: Consist Total Length Error	Difference between the Calculated Train Length and the Consist Total Train Length is greater than expected based on train dynamics	Difference between the Calculated Train Length and the Consist Total Train Length is less than or equal to what is expected based on train dynamics	PTC does not change state	No		EOT, HOT	Yes	No
0x085D [L2I474.085D]	CPUx	CPUx: Reduced speed restriction outside Country of Operation Canada	Track data indicates a speed restriction of type reduced speed on the train's calculated	Sixty (60) seconds have elapsed since the fault was raised	PTC Slice State goes FAILED	No		None	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
			route in a subdivision/ district with a country of operation other than Canada							
0x0860 [L2I474.0860]	CPUx	CPUx: Cab signal on track with Country of Operation Canada	A track data file is loaded and the country of operation for the subdivision/ district is Canada and the track data contains track with a cab signal type of route based or speed enforcement	Sixty (60) seconds have elapsed since the fault was raised	PTC Slice State goes FAILED	No	This fault does not represent a problem with the on-board segment. This fault is raised to alert the back office to an invalid track data file configuration.	None	Yes	No
0x0861 [L2I474.0861]	CPUx	CPUx: Yard limits on track with Country of Operation Canada	A track data file is loaded and the country of operation for the subdivision/ district is Canada and the track data contains track with a flag indicating yard limits are in effect	Sixty (60) seconds have elapsed since the fault was raised	PTC Slice State goes FAILED	No	This fault does not represent a problem with the on-board segment. This fault is raised to alert the back office to an invalid track data file configuration.	None	Yes	No

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Fault Code	Fault Source	Reported Fault	Conditions to Trigger	Conditions to Clear	System Response	Applies Brakes*	Notes	Potential Components to Replace	Report to Office	Prevents Initialization
0x0862 [L21474.0862]	CPUx	CPUx: Track circuit on track with Country of Operation Canada	A track data file is loaded and the country of operation for the subdivision/district is Canada and the track data contains a signal of type "track circuit" or "OS track circuit"	Sixty (60) seconds have elapsed since the fault was raised	PTC Slice State goes FAILED	No	This fault does not represent a problem with the on-board segment. This fault is raised to alert the back office to an invalid track data file configuration.	None	Yes	No

* - PTC will only apply brakes due to a fault when that fault is in the 'Active' or 'Intermittent Active' states.

Note: The following fault codes are obsolete:

0x0320, 0x0321, 0x0322, 0x0323, 0x0324, 0x052A, 0x052B, 0x0534, 0x053A, 0x0542, 0x0543, 0x0544, 0x0550, 0x0552, 0x0557, 0x0570, 0x0571, 0x0580, 0x0581, 0x0582, 0x0583, 0x0590, 0x05B0, 0x05B1, 0x05B2, 0x0600, 0x0610, 0x0611, 0x0612, 0x0613, 0x0615, 0x0616, 0x0624, 0x0627, 0x063B, 0x063C, 0x0642, 0x0643, 0x0701, 0x0742, 0x0744, 0x0745, 0x0746, 0x0804, 0x0813, and 0x0814.

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/// 8 I-ETMS Event Codes

Event Code	Event Description	Conditions to Trigger	Conditions to Clear	Associated Scenario
0x1001	Track location on a different route selected	The “Yes” key is pressed at the “You selected <track solution>. Is this correct?” prompt and the locomotive track location is not already on that <track solution>.	See Note A	DIR01
0x1002	Unmapped track at end of calculated route	The leading edge of the train is within calculated position uncertainty of the end of the train’s calculated route and the train’s calculated route ends at an unmapped track linkage.	See Note A	LOC01
0x1003	Switch aligned to unmapped track	The leading edge of the train is within calculated position uncertainty of the end of the train’s calculated route and the train’s calculated route ends at unmapped track connected to the last switch.	See Note A	LOC02
0x1004	Track not loaded	The leading edge of the train is within calculated position uncertainty of the end of the train’s calculated route and the train’s calculated route ends at a linkage to a subdivision/district for which track is not loaded.	See Note A	LOC03
0x1005	Maximum cross track error	The “Acknowledge” key is pressed at the “Position unknown. Press key to acknowledge that the system will be disabling enforcement” prompt.	See Note A	LOC07, LOC08, LOC09
0x1006	Trailing edge of train exits PTC track	The trailing edge of the train moves from PTC track to non-PTC / PTC entry track and there is no synchronized, non-suspended PTC track under the train or on the train’s calculated route in advance of the train.	See Note A, B	PTC01
0x1007	Different timetable direction selected	The “East”, “West”, “North” or “South” key is pressed at the “What timetable direction would your train travel with the reverser in the forward position?” prompt and the selected calculated direction is different than the current calculated direction of movement.	See Note A	DIR02
0x1008	Locomotive orientation changed	The train orientation changes (forward -> reverse or reverse -> forward) due to a change in the locomotive orientation (front -> back or back -> front) in the summary consist.	See Note A	DIR03
0x1009	Unknown facing switch	The leading edge of the train is within calculated position uncertainty of a facing switch with unknown or unknown restricting status.	See Note A	LOC05
0x100A	Shove behind unknown facing switch	The train orientation is reverse and the locomotive track location is within calculated position uncertainty of a facing switch with unknown status and that switch was under the train when the locomotive track location and calculated direction of movement became known.	See Note A	LOC06
0x100B	Maximum position uncertainty	The “Acknowledge” key is pressed at the “No GPS signal. Press key to acknowledge that the system will be disabling enforcement” prompt.	See Note A	LOC10
0x100C	Farther than allowable rollback	The “Acknowledge” key is pressed at the “Direction unknown. Press key to acknowledge that the system will be disabling enforcement” prompt.	See Note A	LOC11
0x100D	Rollback into exclusion zone	There is an exclusion zone on the train’s calculated route and the leading edge of the train is closer to the start of the train’s calculated route than the switch for which the exclusion zone was generated.	See Note A	LOC12

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Event Code	Event Description	Conditions to Trigger	Conditions to Clear	Associated Scenario
0x100E	Train orientation change in exclusion zone	The train orientation is reverse and there is a facing switch with unknown orientation on the train's calculated route: - under the train or - within calculated position uncertainty behind the trailing edge of the train or - within calculated position uncertainty in advance of the leading edge of the train.	See Note A	LOC13
0x100F	Currently occupied subdivision / district out of sync	The "Acknowledge" key is pressed at the "Synchronization error. Press key to acknowledge that the system will be disabling enforcement" prompt and any subdivision / district occupied by any portion of the train is non-synchronized.	See Note A	SYN01
0x1010	Subdivision / district ahead out of sync	The "Acknowledge" key is pressed at the "Synchronization error. Press key to acknowledge that the system will be disabling enforcement" prompt and no subdivision / district occupied by any portion of the train is non-synchronized.	See Note A	SYN02
0x1011	Train orientation change	The train orientation changes (forward -> reverse or reverse -> forward) due to a change in reverser handle position and - the previous route contained synchronized, non-suspended PTC track in advance of the train and - the current route does not contain synchronized, non-suspended PTC track in advance of the train	See Note A, B	PTC02
0x1012	Switch leading to PTC track alignment change	The entire train is on non-PTC / PTC entry track or non-synchronized PTC track or suspended PTC track and - the train's calculated route is updated not due to a change in train orientation and - the previous route contained synchronized, non-suspended PTC track in advance of the train and - the current route does not contain synchronized, non-suspended PTC track in advance of the train and - the current route does not end at a linkage to a subdivision/district for which track is not loaded	See Note A, B	PTC03
0x1013	Track file discarded	The entire train is on non-PTC / PTC entry track or non-synchronized PTC track or suspended PTC track and - the train's calculated route is updated not due to a change in train orientation and - the previous route contained synchronized, non-suspended PTC track in advance of the train and - the current route does not contain synchronized, non-suspended PTC track in advance of the train and - the current route ends at a linkage to a subdivision/district for which track is not loaded	See Note A, B	PTC04
0x1014	RESERVED			

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Event Code	Event Description	Conditions to Trigger	Conditions to Clear	Associated Scenario
0x1015	Trailing edge of train enters suspended track	The trailing edge of the train enters suspended track and there is no synchronized, non-suspended PTC track under the train or on the train's calculated route in advance of the train.	See Note A, B	PTC05
0x1016	Trailing edge of train exits synchronized PTC track	The trailing edge of the train moves from synchronized track to non-synchronized track and there is no synchronized, non-suspended PTC track under the train or on the train's calculated route in advance of the train.	See Note A, B	SYN03
0x1017	PTC suspension bulletin activated	The "Confirmed" key is pressed at the "Press key to confirm PTC suspension" prompt or warning prompt	See Note A	PTC06
0x1018	Braking fault occurs	The "Acknowledge" key is pressed at the "Braking distance calculation error. Press key to acknowledge that the system will be disabling enforcement" prompt.	See Note A	FLT01
0x1019	Power Cycle	The on-board segment boots up and enters the Self Test state	See Note C	PWR01
0x101A	Approach to trailing switch inconsistent with switch status	The on-board segment state is ACTIVE and - the train orientation is forward and - there is a trailing switch within the calculated position uncertainty in advance of the leading edge of the train and - the calculated route indicates a normal approach to that switch and - the position of that switch is reverse and - that switch is not a spring reverse switch and - that switch is not located inside or coterminous with the limits of an exclusion zone OR The on-board segment state is ACTIVE and - the train orientation is forward and - there is a trailing switch within the calculated position uncertainty in advance of the leading edge of the train and - the calculated route indicates a reverse approach to that switch and - the position of that switch is normal and - that switch is not a spring normal switch and - that switch is not located inside or coterminous with the limits of an exclusion zone	See Note A	LOC15
0x101B	Unknown trailing switch under train without exclusion zone	The on-board segment state is ACTIVE and - the train orientation is forward and - there is a trailing switch with unknown status on the train's calculated route between the calculated position uncertainty in advance of the leading edge of the train and the calculated position uncertainty behind the trailing edge of the train and - that switch is not located inside or coterminous with the limits of an	See Note A	LOC16

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Event Code	Event Description	Conditions to Trigger	Conditions to Clear	Associated Scenario
		exclusion zone		
0x101C	Init or cut in canceled: Config reason C1	The "Cancel" key is pressed at the "System configuration is not compatible" banner with reason text "C1: No common preferred or acceptable software version exists".	See Note A	N/A
0x101D	Init canceled: Config reason C2	The "Cancel" key is pressed at the "Please wait for software download" banner with reason text "C2: No common preferred or acceptable software version is distributed onboard; downloading files".	See Note A	N/A
0x101E	Init canceled: Config reason C3	The "Cancel" key is pressed at the "Please wait for software download" banner with reason text "C3: Distributing preferred or acceptable software version onboard".	See Note A	N/A
0x101F	Init or cut in canceled: Config reason C4	The "Cancel" key is pressed at the "System configuration is not compatible" banner with reason text "C4: No common preferred or acceptable common configuration version exists".	See Note A	N/A
0x1020	Init or cut in canceled: Config reason C5	The "Cancel" key is pressed at the "System configuration is not compatible" banner with reason text "C5: Common preferred or acceptable common configuration version(s) not compatible with current software version".	See Note A	N/A
0x1021	Init or cut in canceled: Config reason C6	The "Cancel" key is pressed at the "Please wait for software download" banner with reason text "C6: No common preferred or acceptable common configuration version compatible with current software is distributed onboard; downloading files".	See Note A	N/A
0x1022	Init or cut in canceled: Config reason C7	The "Cancel" key is pressed at the "Please wait for software download" banner with reason text "C7: Distributing preferred or acceptable common configuration onboard".	See Note A	N/A
0x1023	Init or cut in canceled: Config reason C8 - operating	The "Cancel" key is pressed at the "System configuration is not compatible" banner with reason text "C8: No preferred or acceptable railroad configuration version exists for <railroad>".	See Note A	N/A
0x1024	Init or cut in canceled: Config reason C9 - operating	The "Cancel" key is pressed at the "System configuration is not compatible" banner with reason text "C9: Preferred or acceptable railroad configuration version(s) for <railroad> not compatible with current software version".	See Note A	N/A
0x1025	Init or cut in canceled: Config reason C10 - operating	The "Cancel" key is pressed at the "Please wait for software download" banner with reason text "C10: No preferred or acceptable railroad configuration version for <railroad> is distributed onboard; downloading files".	See Note A	N/A
0x1026	Init or cut in canceled: Config reason C11 - operating	The "Cancel" key is pressed at the "Please wait for software download" banner with reason text "C11: Distributing preferred or acceptable railroad specific configuration onboard for <railroad>".	See Note A	N/A
0x1027	Cut in canceled: Config reason C12	The "Cancel" key is pressed at the "System configuration is invalid" banner with reason text "C12: Current software version is not preferred or acceptable"	See Note A	N/A
0x1028	Init or cut in canceled: Config reason C8 – employing	The "Cancel" key is pressed at the "Logon configuration is not compatible" banner with reason text "C8: No preferred or acceptable railroad configuration version exists for <railroad>".	See Note A	N/A
0x1029	Init or cut in canceled: Config reason C9 -	The "Cancel" key is pressed at the "Logon configuration is not compatible" banner with reason text "C9: Preferred or acceptable railroad configuration version(s) for	See Note A	N/A

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Event Code	Event Description	Conditions to Trigger	Conditions to Clear	Associated Scenario
	employing	<railroad> not compatible with current software version".		
0x102A	Init or cut in canceled: Config reason C10 - employing	The "Cancel" key is pressed at the "Please wait for logon configuration download" banner with reason text "C10: No preferred or acceptable railroad configuration version for <railroad> is distributed onboard; downloading files".	See Note A	N/A
0x102B	Init or cut in canceled: Config reason C11 - employing	The "Cancel" key is pressed at the "Please wait for logon configuration download" banner with reason text "C11: Distributing preferred or acceptable railroad specific configuration onboard for <railroad>".	See Note A	N/A
0x1D01	Diagnostic Code D01	D01 diagnostic code generated from fault(s)	See Note A	N/A
0x1D02	Diagnostic Code D02	D02 diagnostic code generated from fault(s)	See Note A	N/A
0x1D03	Diagnostic Code D03	D03 diagnostic code generated from fault(s)	See Note A	N/A
0x1D04	Diagnostic Code D04	D04 diagnostic code generated from fault(s)	See Note A	N/A
0x1D05	Diagnostic Code D05	D05 diagnostic code generated from fault(s)	See Note A	N/A
0x1D06	Diagnostic Code D06	D06 diagnostic code generated from fault(s)	See Note A	N/A
0x1D07	Diagnostic Code D07	D07 diagnostic code generated from fault(s)	See Note A	N/A
0x1D08	Diagnostic Code D08	D08 diagnostic code generated from fault(s)	See Note A	N/A
0x1D09	Diagnostic Code D09	D09 diagnostic code generated from fault(s)	See Note A	N/A
0x1D10	Diagnostic Code D10	D10 diagnostic code generated from fault(s)	See Note A	N/A
0x1D11	Diagnostic Code D11	D11 diagnostic code generated from fault(s)	See Note A	N/A
0x1D12	Diagnostic Code D12	D12 diagnostic code generated from fault(s)	See Note A	N/A
0x1D13	Diagnostic Code D13	D13 diagnostic code generated from fault(s)	See Note A	N/A
0x1D14	RESERVED			
0x1D15	RESERVED			
0x1D16	RESERVED			
0x1D17	Diagnostic Code D17	D17 diagnostic code generated from fault(s)	See Note A	N/A
0x1D18	Diagnostic Code D18	D18 diagnostic code generated from fault(s)	See Note A	N/A
0x1D19	Diagnostic Code D19	D19 diagnostic code generated from fault(s)	See Note A	N/A
0x1D20	Diagnostic Code D20	D20 diagnostic code generated from fault(s)	See Note A	N/A
0x1D21	Diagnostic Code D21	D21 diagnostic code generated from fault(s)	See Note A	N/A
0x1D22	Diagnostic Code D22	D22 diagnostic code generated from fault(s)	See Note A	N/A
0x1D23	RESERVED			
0x1D24	Diagnostic Code D24	D24 diagnostic code generated from fault(s)	See Note A	N/A
0x1D25	Diagnostic Code D25	D25 diagnostic code generated from fault(s)	See Note A	N/A
0x1D26	Diagnostic Code D26	D26 diagnostic code generated from fault(s)	See Note A	N/A
0x1D27	RESERVED			
0x1D28	Diagnostic Code D28	D28 diagnostic code generated from fault(s)	See Note A	N/A
0x1D29	Diagnostic Code D29	D29 diagnostic code generated from fault(s)	See Note A	N/A
0x1D30	Diagnostic Code D30	D30 diagnostic code generated from fault(s)	See Note A	N/A

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Event Code	Event Description	Conditions to Trigger	Conditions to Clear	Associated Scenario
0x1D31	Diagnostic Code D31	D31 diagnostic code generated from fault(s)	See Note A	N/A
0x1D32	Diagnostic Code D32	D32 diagnostic code generated from fault(s)	See Note A	N/A
0x1D33	Diagnostic Code D33	D33 diagnostic code generated from fault(s)	See Note A	N/A
0x1D34	Diagnostic Code D34	D34 diagnostic code generated from fault(s)	See Note A	N/A

Notes:

- A. The events that reference this note will be cleared after a configurable period calculated as: TBC213 + one (1) seconds. This allows time for the event to be included in a Locomotive Fault Report (02087) message.
- B. Suspended track is track covered by a maximum authorized speed PTC suspension target. Non-suspended track is track which is not covered by a maximum authorized speed PTC suspension target.
- C. The event that references this note will be cleared one (1) second after the on-board segment state transitions from Self Test to Cut Out or Failed. This allows time for this event to be included in a Locomotive Fault Summary Report (02081) message.

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/// 9 I-ETMS Disengagement, Cut Out, and Failure Scenarios

Scenario Code	Scenario Description	Associated Event Code or Fault Code	Notes
FLT01	A braking calculation fault is raised	0x1018	
DIR01	The crew selects a controlling locomotive track location on a different track, possibly to correct an error in their initial selection or in selecting a switch position	0x1001	
DIR02	The crew selects a different timetable direction, possibly to correct an error in their initial selection	0x1007	
DIR03	A summary consist update is received with a change in the locomotive orientation (front -> back or back -> front)	0x1008	
LOC01	At least part of the train is on synchronized, non-suspended PTC track and the train's calculated route ends at unmapped track at the end of a track segment	0x1002	B
LOC02	At least part of the train is on synchronized, non-suspended PTC track and the train's calculated route ends at unmapped track at a switch leg where the other switch leg leads to more mapped track	0x1003	B
LOC03	At least part of the train is on synchronized, non-suspended PTC track and the train's calculated route ends at track that is not loaded	0x1004	B
LOC04	RESERVED		
LOC05	The crew fails to indicate the position of a switch with unknown status and the train passes over that switch	0x1009	
LOC06	The train becomes located on a switch with unknown status and then shoves back over the switch far enough that the on-board segment cannot determine which leg of the switch the locomotive is on	0x100A	
LOC07	The train drives off mapped track onto unmapped track which diverges from the calculated route at a switch not represented in the track data	0x1005	A
LOC08	The on-board segment cannot match erroneous but "valid" GPS to the train's calculated route, possibly due to an obstruction near the track	0x1005	A
LOC09	The crew selects the wrong switch position at an unmonitored or uncommunicating monitored switch and the train's actual route diverges from the calculated route	0x1005	A
LOC10	The on-board segment dead reckons until the maximum calculated position uncertainty is reached	0x100B	
LOC11	The train rolls back farther than the allowable rollback distance	0x100C	
LOC12	There is an exclusion zone under the train and the train rolls back far enough that the leading edge of the train is behind the switch for which the exclusion zone was created	0x100D	
LOC13	At least part of the train is on synchronized, non-suspended PTC track with an exclusion zone under part of the train and the train orientation changes (due to reverser change)	0x100E	B
LOC14	No speed source is available	0x0601, 0x0603, 0x0604, 0x0605, 0x0606, 0x0607, 0x0608, 0x0609, 0x060A, 0x060B, 0x0683	

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Scenario Code	Scenario Description	Associated Event Code or Fault Code	Notes
LOC15	The crew chooses the wrong location during track selection and then moves the train toward a trailing switch aligned for the actual train location	0x0101A	
LOC16	A trailing switch with unknown status is located under the train after the on-board segment state becomes ACTIVE and the train orientation is "forward", for example due to: • A consist update that extends the train length over the switch, or • The train rolling back over the switch or • The GPS drifting over the switch or • The train moving forward over a switch without answering the prompt to verify switch alignment	0x0101B	
SYN01	The on-board segment sets the subdivision/district occupied by the train to non-synchronized	0x100F	
SYN02	With the entire train on non-PTC/PTC entry track or non-synchronized PTC track or suspended PTC track and synchronized, non-suspended PTC track in a different subdivision/district in advance of the train, the on-board segment sets the subdivision/district in advance of the train to non-synchronized	0x1010	B
SYN03	The entire train exits a synchronized PTC subdivision/district and there is no synchronized, non-suspended PTC track in advance of the train	0x1016	B
PTC01	The entire train exits PTC track with no synchronized, non-suspended PTC track ahead	0x1006	B
PTC02	The train orientation changes with the entire train on non-PTC/PTC entry track or non-synchronized PTC track or suspended PTC track with synchronized, non-suspended PTC track in advance of the train in the original direction and no PTC track in advance of the train in the new direction	0x1011	B
PTC03	With the entire train on non-PTC/PTC entry track or non-synchronized PTC track or suspended PTC track, a switch in advance of the train changes position such that synchronized, non-suspended PTC track that was on the calculated route in advance of the train is no longer on the calculated route	0x1012	B
PTC04	With the entire train on non-PTC/PTC entry track or non-synchronized PTC track or suspended PTC track and synchronized, non-suspended PTC track in advance of the train, the on-board segment discards the track file containing that synchronized, non-suspended PTC track	0x1013	B
PTC05	The entire train enters a maximum authorized speed PTC suspension target and there is no synchronized, non-suspended PTC track in advance of the train	0x1015	B
PTC06	The on-board segment generates a maximum authorized speed PTC suspension target over the entire train and there is no synchronized, non-suspended PTC track in advance of the train	0x1017	B
PWR01	The on-board segment is powered off while in a controlling state and then powered back on	0x1019	

Notes:

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- A. Additional post analysis is required to differentiate between disengagement scenarios LOC07, LOC08, and LOC09.
- LOC07: The disengagement takes place near a switch not represented in the track data
 - LOC08: The disengagement takes place near a known GPS obstruction
 - LOC09: The disengagement takes place near a switch for which a switch leg key has been pressed at the “Indicate alignment of switch <switch name>” prompt or “Switch position is unknown indicate alignment of switch <switch name>” warning prompt.
- B. Suspended track is track covered by a maximum authorized speed PTC suspension target. Non-suspended track is track which is not covered by a maximum authorized speed PTC suspension target.

9.1 Recommended Steps for Categorizing Disengagements

When the on-board segment state changes from Active to Disengaged, as indicated by the generation of a Locomotive Position Report (02080) message with Locomotive State 3 – Disengaged after a previous Locomotive Position Report (02080) message with Locomotive State 5 – Active:

FLT01:

If a Locomotive Fault Report (02087) message is generated within TBC213 seconds of the Locomotive Position Report (02080) message with Locomotive State 3 – Disengaged and that Locomotive Fault Report (02087) message contains the vent code listed for the FLT01 disengagement scenario in section 9, then categorize the disengagement as Disengagement Scenario FLT01.

SYN## or PTC##:

If the disengagement remains uncategorized by the previous steps and the Locomotive Position Report (02080) message with Locomotive State 3 – Disengaged contains a known track location and direction of travel and a Locomotive Fault Report (02087) message is generated within TBC213 seconds of that Locomotive Position Report (02080) message and that Locomotive Fault Report (02087) message contains an event code listed for one of the SYN## or PTC## disengagement scenarios in section 9, then categorize the disengagement as the corresponding Disengagement Scenario.

DIR##:

If the disengagement remains uncategorized by the previous steps and the Locomotive Position Report (02080) message with Locomotive State 3 – Disengaged does not contain a known track location and direction of travel and a Locomotive Fault Report (02087) message is generated within TBC213 seconds of that Locomotive Position Report (02080) message and that Locomotive Fault Report (02087) message contains an event code listed for one of the DIR## disengagement scenarios in section 9, then categorize the disengagement as the corresponding Disengagement Scenario.

LOC14:

If the disengagement remains uncategorized by the previous steps and the Locomotive Position Report (02080) message with Locomotive State 3 – Disengaged does not contain a known track location and direction of travel and a Locomotive Fault Report (02087) message is generated within TBC213 seconds of that Locomotive Position Report (02080) message and that Locomotive Fault Report (02087) message contains any of the active or intermittent faults listed for the LOC14 disengagement scenario in section 9, then categorize the disengagement as Disengagement Scenario LOC14.

LOC01 – LOC06, LOC10 – LOC13:

If the disengagement remains uncategorized by the previous steps and the Locomotive Position Report (02080) message with Locomotive State 3 – Disengaged does not contain a known track location and direction of travel and a Locomotive Fault Report (02087) message is generated within TBC213 seconds of that Locomotive Position Report (02080) message and that Locomotive Fault Report (02087) message contains an event code listed for one of

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the LOC01, LOC02, LOC03, LOC05, LOC06, LOC10, LOC11, LOC12, or LOC13 disengagement scenarios in section 9, then categorize the disengagement as the corresponding Disengagement Scenario.

LOC07, LOC08, and LOC09:

If the disengagement remains uncategorized by the previous steps and the Locomotive Position Report (02080) message with Locomotive State 3 – Disengaged does not contain a known track location and direction of travel and a Locomotive Fault Report (02087) message is generated within TBC213 seconds of that Locomotive Position Report (02080) message and that Locomotive Fault Report (02087) message contains event code 0x1005, then categorize the disengagement as Disengagement Scenario LOC07, LOC08, or LOC09, utilizing the post analysis described in the notes in section 9.

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/// Appendix A Configuration Code to Component Code Mapping

Component Configuration on ▼																																
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	X	X	X	X	X	X	X	X	X	X	X	X			X																X	X
2	X	X	X	X	X	X	X	X	X	X					X																X	X
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77	X	X	X	X						X	X	X																				

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78	X	X	X	X						X	X	X																				
79	X	X	X	X						X	X	X																				
80	X	X	X	X						X	X	X																				
81	X	X	X	X						X	X	X																				
82																									X							
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89																									X							
90																															X	X
91																															X	X
92																															X	X
93																															X	X